Random Variable

- Studying Variation pattern in the variable under study

- Central Tendency - A central value around which most of the dataset is clustering.
- Dispersion ← Scatterness
- Skewness ← Departure from Symmetry
- kurtosis ⇒ Peakedness.

Ex:- Tossing a coin Twice - E

$\Omega = \{HH, HT, TH, TT\}$

A = Getting 2 Heads. $P(A) = 1/4$

B = exactly 1 Heads

X = No of Heads.

X	P(X=x)
0	1/4
1	1/2
2	1/4

Random variable (R.V) is a real valued function defined on sample space with some specific Probability.

RV < Discreate RV / continuous RV.

Discrete Random variable

↳ If takes some isolated value with sp. probability.

Sample	Population	
variable (x)	R.V (X)	Parameter (Unknown)
Sample Mean ($\bar{x}$)	Pop. Mean (μ)	
Sample Median ($\tilde{x}$)	Pop. Median ($\tilde{\mu}$)	
Sample Mode ($\hat{x}$)	Pop. Mode ($\hat{\mu}$)	

Statistics ↓ calculated

Rel. freq def^n of Prob:-

x	Freq	Rel. freq.
$L_1 - L_2$	f_1	f_1/n
$L_2 - L_3$	f_2	f_2/n
⋮	⋮	⋮
$L_{k-1} - L_k$	f_k	f_k/n
	n	

Population size = N

Sample size = n

Law of Statistical Regularity say that $\lim_{n \to \infty} \frac{f_i}{n} \to$ Real No $P(L_i \le X \le L_{i+1})$

$n \to$ Population.

Rel Freq Tends to. $\to 1/4$

X = No. of heads

	P $(X = x)$
0	1/4
1	1/2
2	1/4

↓

Prob. dist^n.

x	Freq
0	f_0
1	f_1
2	f_2
	50.

Rel. Freq.

$f_0/50 \to 50 \uparrow \to 1/4$

$f_1/50 \to 1/2$

$f_2/50 \to 1/4$

↑ the Sample size

↓

Sample freq distribution.

DRV :-

Statistical Regularity → if $n \to \infty$ then $\lim_{n\to\infty} \frac{f}{n} \to p_i$.

if pop^n size = N.

Sample size = n.

on the basis of sample data we construct a freq Table

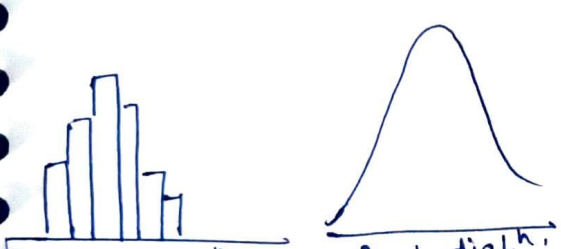
$$\text{Rel. freq} = \frac{f}{n}$$

Objective of Statistics is to study the pattern of variation in the variable under study.

Tabular representation → Freq. Distribution.

Graphical representation → Histogram, Bar diagram.

Measure of different characteristics - central Tendency
- Dispersion
- skewness
- Peakedness



freq dist^n (Sample). $\xrightarrow[n\uparrow]{}$ Prob dist^n.

Sample	Population
Rel freq $\xrightarrow[n\to\infty]{}$	Prob.
$\bar{x} = \frac{\sum f_i x_i}{n} \xrightarrow[n\to\infty]{}$	$\sum_x x P(X=x) = \mu$
$s^2 = \frac{\sum (x_i - \bar{x})^2 f_i}{n}$	$\sum (x-\mu)^2 P(X=\bar{x})$ σ^2
Median $\tilde{x}$	$\tilde{\mu} \to P(X \le \tilde{\mu}) = \frac{1}{2}$
Mode $\bar{x}$	$\bar{\mu} \to \max_x P(X=x) = P(X=\bar{\mu})$

DRV ~~is~~ takes some isolated value with some specific Prob.

Say

$P(X=x)$ $x = \dots$

$P(X=x) = f(x)$ is called P.M.F (Prob Mass fu^n)

E ⇒ Tossing a coin twice

$\Omega = [HH, HT, TH, TT]$.

X = No. of heads

X	P(X=x)
0	1/4
1	1/2
2	1/4
	1

Property of PMF:-

1.) $f(x) \geq 0 \quad \forall x$

2.) $\sum_{x} f(x) = 1$

$x \rightarrow$ Sample value

$X \rightarrow$ Random variable.

$\psi(x) \rightarrow$ fun of x.

Expectation $\rightarrow E(\psi(x))$ = is defined as.

$$E(\psi(x)) = \sum_{x} \psi(x) \cdot P(X=x)$$

(I) $$E(X) = \sum_{x} x \cdot P(X=x) = \mu$$

$$\sigma^2 = E(X-\mu)^2 = E(X^2 - 2\mu X + \mu^2)$$

$$= E(X^2) - 2\mu^2 + \mu^2$$

$$= E(X^2) - \mu^2$$

(II) $$\sigma^2 = E(X^2) - (E(X))^2$$

X	P(X=x)	cum. Prob.
0	1/4	1/4
(1)	1/2 →	3/4 → 1/2 is lying here.
2	1/4	1

Median → (1)

Mode ⇒ Highest $(\frac{1}{4}, (\frac{1}{2}), \frac{1}{4})$

Mode = 1

(i) Raw Moment (r^{th} order) (μ_r')

$$\mu_r' = E(X^r) = \sum x^r P(X=x)$$

(ii) r^{th} order central Moment (μ_r)

$$\mu_r = E(X-\mu)^r$$

$$= \sum_x (x-\mu)^r P(X=x)$$

Mean $= \mu_1'$
variance $= \sigma^2 = \mu_2$
Skewness $= \frac{\mu_3^2}{\mu_2^3}$
kurtosis $= \frac{\mu_4}{\mu_2^2}$

Measure of Skewness $= \frac{\mu_3^2}{\mu_2^3}$

Sample Raw Moment $= m_r' = \frac{1}{n} \sum x_i^r f_i$

Sample central Moment $= m_r = \frac{1}{n} \sum (x_i - \bar{x})^r f_i$

(iii) Factorial Moment ($\mu_{[r]}$)

$$\mu_{[r]} = \sum_x {}^x P_r \cdot P(X=x) = E({}^X P_r)$$

$$= \sum \frac{x_i}{(x-r)!} P(X=x)$$

$$\sum_x x(x-1)\cdots(x-r+1) \cdot P(X=x)$$

Ex- Express μ_r in terms of μ_r'.

$$\mu_r = E(x-\mu)^r$$

$$= \sum_x (x-\mu)^r \cdot P(X=x)$$

Expend Binomial expansion.

Prob:- Consider, a Prob. r.v X with Prob dist

$f(x) = Kx, \quad x = 1, 2, \ldots, n.$

(i) Find out K.
(ii) Find out μ.
(iii) Find out σ^2.
(IV) Find out Median.
(V) Find out Q_1 & Q_3.

$Q_1 \rightarrow$ First Quartile
$Q_3 \rightarrow$ Third Quartile.
$Q_2 \rightarrow$ Median.

(i) $\sum_x f(x) = 1$.

$K \sum x = 1$

$K\left(\frac{n(n+1)}{2}\right) = 1$

$K = \frac{2}{n(n+1)}$

$P(X = x) = f(x)$.

(ii) $\mu = \sum x \cdot P(X = x)$

$= \sum x \cdot K \cdot x$.

$= K \sum_1^n x^2$

$= \frac{2}{n(n+1)} \left[\frac{n(n+1)(2n+1)}{6\ 3}\right] = \left(\frac{2n+1}{3}\right)$

(III) $\sigma^2 = E(X^2) - \mu^2$

$= \sum x^2 \cdot P(X = x) - \left(\frac{2n+1}{3}\right)^2$

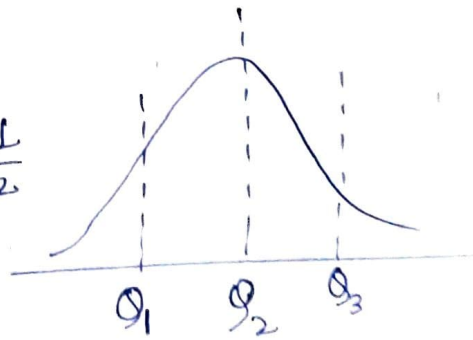
$= K \sum x^3 - \left(\frac{2n+1}{3}\right)^2 = K\left(\frac{n(n+1)}{2}\right)^2 - \left(\frac{2n+1}{3}\right)^2$

(IV) $P(x \le \tilde{\mu}) = \frac{1}{2}$

$\sum_1^{\tilde{\mu}} Kx = \frac{1}{2}$

$K\left(\frac{\tilde{\mu}(\tilde{\mu}+1)}{2}\right) = \frac{1}{2}$

$\tilde{\mu} = \checkmark$.



(V) $P(x \le Q_1) = 0.25$

$P(x \le Q_3) = 0.75$

(I) Bernoulli's Trial

Let us consider some trials with the following Property.

1.) The Trials are indep.

2.) Each Trial results in 2 possible outcome < Success / Failure

3.) Success Prob. P is fixed for all the trial.

Define a r.v. such that $X_i = \begin{cases} 1 & \text{if } i\text{th Trial results in Success.} \\ 0 & \text{otherwise.} \end{cases}$

X_i	$P_{X_i}(x)$
1	p
0	q

$$E(X_i) = 1 \times p + 0 \cdot q = p.$$

$$\text{Var}(X_i) = E(X_i^2) - (E(X_i))^2$$

$$= p - p^2 = p(1-p) = pq.$$

now consider n such Trials. $X_i = \begin{cases} 1 & \text{Success} \\ 0 & \text{o.w.} \end{cases}$

Define the R.V.

$$X = X_1 + X_2 \text{ ------ } + X_n.$$

= No. of success in n independent Ber. Trials

$X \sim \text{Bin}(n, p)$

X is called Binomial Dist^n with Parameter n & p

$$\boxed{P(X=x) = {}^nC_x \cdot p^x \cdot q^{n-x}} = f(x) = \text{PMF}$$

$x \to 0, 1, 2, \text{ ---- } n.$

(R1) $f(x)$ is a PMF.

$$\sum_x \binom{n}{x} p^x q^{n-x} = (p+q)^n = 1.$$

(R2) $$E(X) = \sum_x x \binom{n}{x} \cdot p^x q^{n-x} = \sum_x \frac{n!}{x(n-x)!(x-1)!} p^x q^{n-x}$$

$$= np \sum \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} q^{(n-1)-(x-1)}$$

$$= np(p+q)^{n-1} = np.$$

(R3) $\text{Var}(X) = E(X-\mu)^2$ $\quad (\mu = np)$

$= E(X^2) - \mu^2$

$= E\big(X(X-1)\big) + E(X) - \mu^2$

$E(X(X-1)) = \sum x(x-1)\binom{n}{x} \cdot p^x \cdot q^{n-x}$

$= \sum \frac{n!}{(x-2)!(n-x)!} p^x \cdot q^{n-x}$

$= n(n-1)\cdot p^2 \sum \frac{(n-2)!}{(x-2)!(n-x)!} p^{x-2} q^{(n-2)-(x-2)}$

$= n(n-1)p^2 (p+q)^{n-2} = n(n-1)p^2$

$\text{Var}(X) = n(n-1)p^2 + \mu - \mu^2 = n^2p^2 - np^2 + np - n^2p^2$

$= np(1-p) = \underline{npq}$.

n = no of trial

p = succen prob.

(R4) Mode of Binomial Distribution:-

$P(X=x) = \binom{n}{x} p^x q^{n-x} \quad x = 0, 1, \ldots n.$

$$\frac{P(X=x)}{P(X=x-1)} = \frac{\binom{n}{x} p^x q^{n-x}}{\binom{n}{x-1} p^{x-1} q^{n-x+1}} = \frac{\frac{n!}{(n-x)!\,x!} \times p \times q^{-1}}{\frac{n!}{(x-1)!\,(n-x+1)!}}$$

$$= \frac{n-x+1}{x} \times \frac{p}{q}$$

$P(X=x) \gtrless P(X=x-1)$

if $(n-x+1)p \gtrless qx$

$np - px + p \gtrless qx$

$p(n+1) \gtrless x$

$\Rightarrow \boxed{x \lessgtr (n+1)p}$.

$P(X=x) > P(X=x-1)$ if $x < (n+1)P$.

$P(X=x) = P(X=x-1)$ if $x = (n+1)P$.

$P(X=x) < P(X=x-1)$ if $x > (n+1)P$.

Case-I $(n+1)P$ is integer.

$m_0 = (n+1)P$.

$X = \underbrace{0, 1, 2, \ldots\ldots m_0-1}_{x<(n+1)P}, \underbrace{m_0}_{x=(n+1)P}, \underbrace{m_0+1 \ldots\ldots n}_{x>(n+1)P}$.

$P(X=0) < P(X=1) \ldots\ldots \underbrace{\leq P(m_0-1) = P(X=m_0)}_{\text{Highest Probability}} > P(X=m_0+1) \ldots\ldots P(X=n)$

Modes $= m_0, m_0-1$

$m_0 = (n+1)P$ is an integer.

Case-II $(n+1)P \rightarrow$ Not an integer

Take $m_0 = [(n+1)P]$ = Greatest integer.

$(n+1)P = 3.7$

$m_0 = 3$.

$X = \underbrace{0, 1, 2, \ldots\ldots (m_0-1)}_{x<(n+1)P}, \underbrace{m_0, m_0+1 \ldots\ldots n}_{x>(n+1)P}$.

$P(X=0) < P(X=1) \ldots\ldots \underbrace{\leq P(X=m_0)}_{\text{Max}^M \text{ Prob.}} > P(X=m_0+1) \ldots\ldots P(X=n)$

Hence Mode $= \underline{m_0} = [(n+1)P]$.

Ex $\Rightarrow n=8, P=\frac{1}{4}$

$(n+1)P = 9 \times \frac{1}{4} = 2.25$.

$m_0 \Rightarrow 2 =$ Mode

Ex:- $(n+1)P$

$n=7, P=\frac{1}{4}$

$(n+1)P = 2 = m_0$

Mode $= 2, 1$

Fitting Binomial Distribution :-

Ex:- 1.) # boys in a class.
2.) # defective in a lot.
3.) # good item in a lot.

when sample size = finite we can fit Bin.

X	observed freq	Rel. Freq	Exp. Freq (N × Rel. Freq)
0	f_0		
1	f_1		
2	f_2		
⋮	⋮		
n	f_n		
	N		

considering Binomial.

$P(X=x) = \binom{n}{x} \hat{p}^x \hat{q}^{n-x}$

$N \times P(X=x) = \text{exp. freq}$

↓ Rel. Freq

Mean $= np$.

$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = np$

$\hat{p} = \frac{\bar{x}}{n}$ = p estimate.

$X \sim Bin(n, p)$ → Expected Frequency

↓

if we consider that the Particular variable is following a r.v. X.

then $N \times P(X=x) \Rightarrow$ exp. freq.

if pattern of variation is matching of obs freq & Exp. freq. our assumption is correct.

1.) Find out $\bar{x}$ freq. dist^n

$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$

2.) Estimate $\hat{p} = \frac{\bar{x}}{n}$

3) Find out $P(X=x) = \binom{n}{x} \cdot \hat{p}^x \cdot \hat{q}^{n-x}$.

4.) " Exp. freq $= N \times P(X=x)$

5.) Compare obs. freq & exp. freq.

$P_{(X=x)} = \frac{n-x+1}{x} \times \frac{p}{q} P(x=x-1)$

using recursive formula.

Example:- Tossing a coin for 10 times. (Baised / Unbaised)

# heads	obs. Freq	$\frac{n-x+1}{x}$	$\left(\frac{n-x+1}{x} \times \frac{\hat{p}}{\hat{q}}\right)$	$P(X=x) = \left(\frac{n-x+1}{x} \times \frac{\hat{p}}{\hat{q}}\right) \times P(x-1)$	exp. freq $N \times P(X=x)$
0	f_0	–	–	$(1-\hat{p})^n$	
1	f_1				
2	f_2				
⋮	⋮				
n=10	f_{10}				
	N			1 (Approx) (Adjust in First / Last value)	

Fitting Binomial Dist:

⇒ checking whether $X \sim Bin(n, P)$

$$E(X) = n\hat{p} = \bar{x}$$

$$P(X=x) = \binom{n}{x} \hat{p}^x \hat{q}^{n-x}$$

$$\bar{x} = \frac{\sum_{i=1}^{10} f_i x_i}{\sum f_i} = n\hat{p}$$

$$\boxed{\hat{p} = \frac{\bar{x}}{n}}$$

if the assumption is true → then expected Rel. Freq

$= P(X=x)$

where $X \sim Bin(n, P)$

expected freq $= N \times P(X=x)$

$$\left[\frac{P(X=x)}{P(X=x-1)} = \frac{n-x+1}{x} \times \left(\frac{p}{q}\right)\right]$$

$$P(X=x) = \frac{n-x+1}{x} \times \frac{p}{q} \times P(X=x-1)$$

if Pattern of variation is matching then our assumption is true. (in exp. freq & obs freq)

(II) Poission Distribution:-

Situation in Binomial Dist^n.

where There are two possible outcomes.

$n \to \infty$ $\quad p \to 0$.

Such that $np = \lambda$ (Moderate)

$$\binom{n}{x} p^x q^{n-x}$$

$\lim_{\substack{n\to\infty \\ p\to 0 \\ np=\lambda}} R$

$$\lim_R \frac{n(n-1)\cdots(n-x+1)}{x!} p^x (1-p)^{n-x}.$$

$$\lim_R \frac{n}{n}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{x-1}{n}\right)(np)^x \frac{(1-p)^n}{(1-p)^x}$$

$$\lim_R \frac{n}{n}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{x-1}{n}\right)\frac{\left(1-\left(\frac{\lambda}{n}\right)\right)^n}{\left(1-\left(\frac{\lambda}{n}\right)\right)^x}$$

$\left(1-\frac{1}{n}\right)$, $\left(1-\frac{2}{n}\right)$, $\left(1-\frac{x-1}{n}\right)$ $\to 1$; $\left(1-\left(\frac{\lambda}{n}\right)\right)^x \to 1$

$$\frac{\lambda^x}{x!} \lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^n$$

$$\frac{\lambda^x}{x!} e^{\frac{-\lambda}{n}\times n} = \frac{e^{-\lambda}\cdot\lambda^x}{x!} \qquad x = 0, 1, \ldots \infty.$$

$\Downarrow$ Poission Distribution.

$$= \lim_R \binom{n}{x} p^x q^{n-x} = \text{Poission Dist}^n.$$

X is said to follow Poission dist^n with λ.

PMF of $\boxed{P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}}$

$x = 0, 1, \ldots \infty$.

(R1) $f(x)$ is a PMF.

$$\sum P(X=x) = \sum_{0}^{\infty} \frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$= e^{-\lambda} \cdot e^{\lambda} = 1.$$

(R2) $$E(X) = \sum_{0}^{\infty} x \cdot e^{-\lambda} \frac{\lambda^{x}}{x!}$$

$X \sim P(\lambda)$.

$$= e^{-\lambda} \cdot \lambda \sum \frac{\lambda^{x-1}}{(x-1)!} = e^{-\lambda} \cdot \lambda \cdot e^{\lambda}.$$

$$= \boxed{\lambda}$$

$\boxed{E(X) = \lambda}$ = Population Mean of the distribution.

(R3) $$Var(X) = E(X^2) - E^2(X)$$

$$= E(X(X-1)) + E(X) - \mu^2.$$

$$E(X(X-1)) = \sum x(x-1) \frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$= \sum \frac{e^{-\lambda} \cdot \lambda^{x}}{(x-2)!}$$

$$= e^{-\lambda}\lambda^2 \sum_{x} \frac{\lambda^{x-2}}{(x-2)!} = e^{-\lambda} \cdot \lambda^2 \cdot e^{\lambda} = \lambda^2$$

$$Var(X) = \lambda^2 + \lambda - \lambda^2$$

$\boxed{Var(X) = \lambda}$

$\boxed{S.D = \sqrt{\lambda}}$

(R4) Mode $$\frac{P(X=x)}{P(X=x-1)} = \frac{\frac{e^{-\lambda}\lambda^{x}}{x!}}{\frac{e^{-\lambda}\lambda^{(x-1)}}{(x-1)!}} = \left(\frac{\lambda}{x}\right)$$

$$P(X=x) \gtreqless P(X=x-1)$$

$$x \lessgtr \lambda$$

$P(X=x) > P(X=x-1)$ if $x < \lambda$.

$P(X=x) = P(X=x-1)$ if $x = \lambda$.

$P(X=x) < P(X=x-1)$ if $x > \lambda$.

Case-I λ is an integer. (λ_0 say)

$P_{(X=1)}$

$P(X=0) < P(X=1) \longleftarrow \cdots P(X=\lambda_0-1) = P(X=\lambda_0) > P(X=\lambda_0+1) \cdots$

Modes $= \lambda_0, \lambda_0 - 1$. $\lambda_0 = \lambda$

Case-II if λ is not an integer

$\lambda_0 = [\lambda] = G.I[\lambda]$.

$X = 0, 1, 2, \cdots \lambda_0, \lambda_0+1 \cdots \infty$

$X < \lambda$ $\quad$ $X > \lambda$.

$P(X=0) < P(X=1) \cdots < P(X=\lambda_0) > P(X=\lambda_0+1) \cdots$

Mode $= \lambda_0$. $\lambda_0 = [\lambda]$

Fitting of Poission Distribution:

- # Accidents/Day
- # Defects/unit area
- # Telephone calls/wrt time
- # Patient arrived/wrt time.

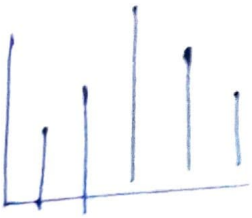
λ = Population Mean

$$\hat{\lambda} = \bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

Fit → $\hat{\lambda}$ estimate.

Till 'k'

$$P(x=x) = \frac{\lambda}{x} P(x=x-1)$$



# defect x	obs. Freq	$\frac{\hat{\lambda}}{x}$	$\frac{\hat{\lambda}}{x} \times P(x=x-1)$	Exp. Freq. $N \times P(x=x)$
0	f_0		$e^{-\hat{\lambda}}$	
1	f_1	$\hat{\lambda}/1$		
2	f_2	$\hat{\lambda}/2$		
⋮	⋮	⋮	⋮	⋮
k	f_k	$\hat{\lambda}/k$		
>K	0			
	N		1 (Approx). (Manage gap in first & Last)	N

→ Compare Pattern of variation Present in exp freq & obs freq

Measure of central Tendency $\Rightarrow \mu = \mu_1'$

" " Dispersion $\Rightarrow \sigma^2 = \mu_2$.

" Skewness $\Rightarrow \frac{\mu_3^2}{\mu_2^3}$ we need to calculate μ_3.

" Kurtosis $\Rightarrow \frac{\mu_4}{\mu_2^2}$ μ_4 need to cal.

Moment Generating function (MGF) :-

$$M_X(t) = E(e^{tx})$$

$$= \sum_x e^{tx} P(X=x)$$

$$= \sum_x \left(1 + \frac{tx}{1!} + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3!} + \cdots\right) P(X=x)$$

$$M_x(t) = 1 + \frac{t}{1!} \sum x \cdot P(x=x) + \frac{t^2}{2!} \sum_x x^2 P(x=x) + \cdots$$

$$= \sum_{r=0}^{\infty} \frac{t^r}{r!} \mu_r'$$

Co-eff of $\frac{t^r}{r!}$ in $M_x(t)$ is μ_r' (r^{th} order raw Moment)

(Provided MGF exist)

$$\boxed{\mu_r' = \frac{d^r}{dt^r} M_x(t) \Big|_{t=0}}$$

Property :- Each Distribution is having a unique MGF if it exists.

Parameters → μ_r —(Formula)→ μ_r' —(diff)→ MGF

if $X \sim Bin(n, p)$

$$M_x(t) = \sum e^{tx} \binom{n}{x} p^x q^{n-x}$$

$$= \sum \binom{n}{x} (pe^t)^x \cdot q^{n-x}$$

$$= (q + pe^t)^n$$

$$E(X) = \mu_1' = \frac{d}{dt} M_x(t) \Big|_{t=0}$$

$$n(q+pe^t)^{n-1}(pe^t) \Big|_{t=0}.$$

$$n \times 1 \times p = \underline{np}$$

$$\mu_2' = \frac{d^2}{dt^2} M_x(t) \Big|_{t=0}$$

$$npe^t(q+pe^t)^{n-1} + n(n-1)(q+pe^t)^{n-2} \times pe^t \times pe^t \Big|_{t=0}.$$

$$np + n(n-1)p^2 = \underline{np + n^2p^2 - np^2}.$$

$$Var(X) = E(X^2) - E^2(X).$$

$$\mu_2' - (\mu_1')^2.$$

$$= np + n^2p^2 - np^2 - n^2p^2$$

$$= np(1-p) = \underline{npq}.$$

Exercise- Cal. Mean & variance for Poission(λ) using $M_x(t)$.

MGF About μ :-

$$M_{(x-\mu)}(t) = E\left(e^{t(x-\mu)}\right)$$

$$= \sum_{x} e^{t(x-\mu)} P(X=x)$$

$$\left[\mu_r = \frac{d^r}{dt^r} M_{x-\mu}(t)\Big|_{t=0}\right]$$

Exercise:- Find out Measure of skewness & kurtosis for $Bin(n,p)$.

μ_3' ⌐
μ_4' ⌐] MGF.

use Relation b/w μ_r & μ_r'.

(III) Geometric Distribution:-

Consider a sequence of Bernoulli's ~~theorem~~ Trial

X = no. of failures preceeding the first success.

Success prob = p.

$X = 0, 1, 2, \ldots\ldots \infty$.

PMF $\Rightarrow$ $P(X=x) = q^x \times p$. $\quad x = 0, 1, 2 \ldots\ldots \infty$.

R-1. $\sum_x P(X=x) = p \sum_0^\infty q^x$

$\Rightarrow \frac{p}{1-q} = 1$

R-2. ~~Mean = μ = $\sum_0^\infty x \cdot P(X=x) = \sum_0^\infty x \times q^x \times p$~~

(Mean & variance of Geometric Distribution)

$X \sim Geo(p)$.

$$\boxed{E(X) = \mu = \frac{q}{p}}$$

$M_x(t) = E(e^{tx})$.

$= \sum_x e^{tx} \cdot q^x \cdot p$.

$= p \sum_x (qe^t)^x$.

$= \frac{p}{1-qe^t}$

$E(X) = \mu = \mu_1' = \frac{d}{dt} M_x(t) \Big|_{t=0}$

$= \frac{-p}{(1-qe^t)^2} \times (-qe^t) \Big|_{t=0}$

$= \frac{pq}{p^2} = \frac{q}{p}$.

$$Var(x) = E(x^2) - E^2(x)$$

$$= \mu_2' - (\mu_1')^2$$

$$\mu_1' = \frac{pq\,e^t}{(1-qe^t)^2} = pq\,e^t(1-qe^t)^{-2}$$

$$\mu_2' = pq\left[e^t(1-qe^t)^{-2} + e^t(+2)(1-qe^t)^{-3}(+qe^t)\right]$$

$$= pq\left[\frac{1}{p^2} + \frac{2q}{p^3}\right]$$

$$= q\left[\frac{1}{p} + \frac{2q}{p^2}\right]$$

$$\mu_2 = \mu_2' - (\mu_1')^2 = \frac{q}{p} + \frac{2q^2}{p^2} - \frac{q^2}{p^2}$$

$$= \frac{q}{p} + \frac{q^2}{p^2} = \frac{qp+q^2}{p^2} = \frac{q(p+q)}{p^2} = \frac{q}{p^2}$$

$$\boxed{Var(x) = \frac{q}{p^2}}$$

Fitting a Geometric Distribution:-

X	obs freq	Exp. ~~freq~~ P(X=x) Prob.	Exp. freq = N×P(X=x)
0	f_0	$\hat{p}$	
1	f_1	$\hat{q}\hat{p}$	
2	f_2	$\hat{q}^2\hat{p}$	$N \times P(X=x)$
⋮	⋮	⋮	
n	f_n	$\hat{q}^n \cdot \hat{p}$	
>n	0	→	
	N	1	

Exp. ~~freq~~ Prob. $= \hat{q}^x \cdot \hat{p}$

$$\bar{x} = \frac{\sum f_i x_i}{N} = \frac{1-p}{p}$$

$$\boxed{\hat{p} = \frac{1}{\bar{x}+1}}$$

(IV) Negative Binomial Distribution:-

Consider a seq. of Bernoulli's Trial.

Define a R.V.

X = no. of failures preceeding r^{th} Success.

$X = 0, 1, \dots \infty$.

F S S F ... S

$(r-1) \to S$

$x \to F$

r^{th} Success

$$P(X=x) = (q^x \cdot p^{r-1} \cdot p)\binom{x+r-1}{r-1}$$

$$= \binom{x+r-1}{x} q^x \cdot p^r \qquad x = 0, 1, \dots$$

(R1) $\sum P(X=x) = p^r \sum_x \binom{x+r-1}{x} q^x$

$= p^r \times (1-q)^{-r}$

$= \frac{p^r}{p^r} = 1$

Mean & Variance:-

MGF of -ve Bin. Distribution (r, p)

$M_x(t) = E(e^{tx})$

$= \sum_x e^{tx} \binom{x+r-1}{x} q^x \cdot p^r$

$= p^r \sum \binom{x+r-1}{x} (qe^t)^x$

$= p^r (1-qe^t)^{-r}$

$E(X) = \mu_1' = \frac{d}{dt}(p^r (1-qe^t)^{-r})$

$= p^r \left[-r(1-qe^t)^{-r-1}(-qe^t) \right]_{t=0}$

$= p^r \times \frac{q}{p} \times r \times p^{-r-1}$

$$\boxed{E(X) = \frac{rq}{p}}$$

$Var(X) = \mu_2' - \mu_1'^2$

$\mu_1' = \frac{d}{dt}\left[qrp^r e^t (1-qe^t)^{-r-1} \right]$

$= qrp^r \left[e^t(1-qe^t)^{-r-1} + e^{t(+r+1)}(1-qe^t)^{-r-2}(qe^t) \right]$

$qrp^r \left[\frac{p}{p^{r+2}} + \frac{(1+r)qe^t}{p^{r+2}} \right]$

$qr \left[\frac{p+q+qr}{p^2} \right] \Rightarrow \frac{(1+qr)qr}{p^2}$

$Var(X) = \frac{(1+qr)qr}{p^2} - \frac{r^2q^2}{p^2} = \frac{qr + q^2r^2 - r^2q^2}{p^2}$

$\Rightarrow \frac{qr}{p^2}$

$$\boxed{Var(X) = \frac{rq}{p^2}}$$

Fitting Neg- Binomial Distribution:-

(r, p).

$\mu = \frac{\sum f_i x_i}{N} = \frac{\hat{r}\hat{q}}{\hat{p}}$

$S^2 = \frac{1}{N-1}\sum (x_i - \bar{x})^2 f_i$

$= \frac{\hat{r}\hat{q}}{\hat{p}^2}$

$\hat{p}, \hat{r} \Rightarrow$ known.

$$\frac{P(x=x+1)}{P(x=x)} = \frac{\binom{x+r}{x+1} p^r q^{x+1}}{\binom{x+r-1}{x} p^r q^x}$$

$$= \left(\frac{x+r}{x+1}\right) q$$

$$P(x=x+1) = \left(\frac{x+r}{x+1}\right) q \times P(x=x)$$

Recursive Relations.

X	Freq	P(X=x) (Exp. Prob)	Exp. Freq N × P(X=x)	$\frac{x+r}{x+1}$	$\left(\frac{x+r}{x+1}\right)\hat{q}$	P(X=x)
0	f_0	From Last column				
1	f_1			✓	✓	
⋮	⋮					
k	f_k					
>k	0		N			1
	N					

Hypergeometric Distribution :-

(V) situation Lot of size N < Type-I / Type-II

No of type-I item = N_1 ⇒ $N_1 + N_2 = N$

II = N_2

Take a sample of size = n.

X = # Type-I items in the sample.

$X = 0, 1, 2, \dots, \min(n, N_1)$

X is said to follow Hypergeometric Distribution with Parameters (N, N_1, n).

$$P(X=x) = \frac{\binom{N_1}{x}\binom{N_2}{n-x}}{\binom{N}{n}} = \frac{\binom{N_1}{x}\binom{N-N_1}{n-x}}{\binom{N}{n}}$$

(R1) $\sum P(X=x) \Rightarrow \sum_x \frac{\binom{N_1}{x}\binom{N_2}{n-x}}{\binom{N}{n}}$ = co-eff of t^{x+n-x} in $(1+t)^{N_1}(1+t)^{N_2}$

$\sum \binom{N_1+N_2}{n} = \binom{N}{n}$

$\binom{N_1}{x}$ → co-eff of t^x in $(1+t)^{N_1}$.

$\binom{N_2}{n-x}$ → " " t^{n-x} in $(1+t)^{N_2}$

(R2) Mean & variance:- (Hypergeometric)

p = Proportion of type-I objects in the lot

$N_1 = NP \qquad q = (1-p)$

$N_2 = Nq$

$$P(X=x) = \frac{\binom{NP}{x}\binom{Nq}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, \ldots, \text{Min}(NP, n)$$

Parameter (N, P, n)

$$\mu_{[r]} = E\left({}^{X}P_r\right) = \sum_x \frac{x!}{(x-r)!} P(X=x)$$

Hypergeometric $\rightarrow N, n, N_1$

$$\mu_{[r]} = \sum_x \frac{x!}{(x-r)!} \frac{\binom{N_1}{x}\binom{N_2}{n-x}}{\binom{N}{n}}$$

$$\frac{1}{\binom{N}{n}} \sum \frac{x!}{(x-r)!} \frac{N_1! \times N_2}{(N_1-x)!\, x! \times (N_2-n+x)!\,(n-x)!}$$

$$\frac{1}{\binom{N}{n}} \sum \frac{N_1!}{(N_1-r)!} \frac{(N_1-r)!}{(x-r)!\,(N_1-x)!} \binom{N_2}{n-x}$$

$$\frac{\binom{N_1 P_r}{}}{\binom{N}{n}} \sum_x \binom{N_1-r}{x-r}\binom{N_2}{n-x}$$

$$\frac{\binom{N_1 P_r}{}}{\binom{N}{n}} \binom{N_1+N_2-r}{n-r}$$

$$\mu_{(r)} = \frac{N_1!}{(N_1-r)!} \times \frac{(N-r)!}{(n-r)!\,(N-n)!} \times \frac{(N-n)!\,n!}{N!}$$

$$\mu_{(r)} = \frac{N_1 P_r \; {}^{n}P_r}{N P_r}$$

$$\boxed{E(x) = \mu_{[1]} = \frac{n N_1}{N}}$$

$$\text{var}(x) = E(x^2) - E^2(x)$$

$$= E(x(x-1)) + E(x) - (E(x))^2$$

$$= \mu_{(2)} + \mu_{(1)} - \mu_{(1)}^2$$

$$\mu_{(2)} = \frac{N_1(N_1-1)(n)(n-1)}{N(N-1)}$$

$$\mu_{(3)} = \frac{N_1 P_3 \times {}^{n}P_3}{N P_3}.$$

$$\text{var}(x) = \frac{N_1 n (N_1-1)(n-1)}{N(N-1)} + \frac{nN_1}{N} - \frac{n^2 N_1^2}{N^2}$$

$$= \frac{N_1 n N (N_1-1)(n-1) + N(N-1) n N_1 - n^2 N_1^2 (N-1)}{N^2(N-1)}$$

$$= \frac{(N_1^2 nN - N_1 nN)(n-1) + nN^2 N_1 - nNN_1 + n^2 N_1^2 - n^2 N_1^2 N}{N^2(N-1)}$$

$$= \frac{N_1^2 n^2 N - N_1 n^2 N - N_1^2 nN + N_1 nN + nN^2 N_1 - nNN_1 + n^2 N_1^2 - n^2 N_1^2 N}{N^2(N-1)}$$

$$\frac{n^2 N_1^2 \; nN^2 N_1 + n^2 N_1^2 - N_1 n^2 N - N_1^2 nN}{N^2(N-1)}$$

$$\Rightarrow \frac{nN_1}{N}\left[\frac{N^2 + nN_1 - nN - N_1 N}{N(N-1)}\right]$$

$$= \frac{nN_1}{N} \frac{(N-n)(N-N_1)}{N(N-1)}$$

$$= \frac{nN_1}{N} \times \frac{N-N_1}{N} \times \frac{N-n}{N-1}$$

$$= \frac{nN_1}{N}\left(1-\frac{N_1}{N}\right)\left(\frac{N-n}{N-1}\right)$$

1st Nov 21

Lot Size = N

Sample Size = n

Type-I object = N_1

$\boxed{N_1 = NP}$

$$P(X=x) = \frac{\binom{N_1}{x}\binom{N-N_1}{n-x}}{\binom{N}{n}} = f(x)$$

$$P(X=x) = \frac{\binom{NP}{x}\binom{NQ}{n-x}}{\binom{N}{n}}$$

---

r^{th} order Factorial Moment :-

$$\mu_{[r]} = \frac{N_1 P_r \; {}^{n}P_r}{N P_r}$$

$$E(X) = \frac{nN_1}{N}$$

$$Var(X) = \frac{nN_1}{N}\left[1-\frac{N_1}{N}\right]\left[\frac{N-n}{N-1}\right] = \mu_{[2]} + \mu_{[1]} - (\mu_{[1]})^2$$

Mode of Hypergeometric Dist^n:-

$$\frac{P_{(X=x)}}{P_{(X=x-1)}} = \frac{\binom{N_1}{x}\binom{N_2}{n-x}\cancel{\binom{N}{n}}}{\cancel{\binom{N}{n}} \times \binom{N_1}{x-1}\binom{N_2}{n-x+1}}$$

$$= \frac{\frac{N_1!}{(N_1-x)!\,x!} \times \frac{N_2!}{(n-x)!\,(N_2-n+x)!}}{\frac{N_1!}{(N_1-x+1)!\,(x-1)!} \times \frac{N_2!}{(N_2-n+x-1)!\,(n-x+1)!}}$$

$$\frac{P_{(X=x)}}{P_{(X=x-1)}} = \frac{(N_1-x+1)(n-x+1)}{x\,(N_2-n+x)}$$

$$P_{(X=x)} \gtreqless P_{(X=x-1)}$$

$$(N_1-x+1)(n-x+1) \gtreqless x(N_2-n+x)$$

$$N_1n - N_1x + N_1 - nx + x^2 - x + n - x + 1 \gtreqless N_2x - nx + x^2$$

$$N_1n + N_1 + n + 1 \gtreqless N_2x + N_1x + 2x$$

$$x(N_2+N_1+2) \lessgtr N_1n + N_1(n+1) + 1(n+1)$$

$$x(N_1+N_2+2) \lessgtr (N_1+1)(n+1)$$

$$\boxed{x \lessgtr \frac{(N_1+1)(n+1)}{(N_1+N_2+2)}}$$

$$\boxed{x \lessgtr \frac{(N_1+1)(n+1)}{(N+2)}}$$

Case-I $\frac{(N_1+1)(n+1)}{(N+2)} = m_0$ is integer $(m_0, m_0-1) \rightarrow$ Mode

Case-II " is not an integer. (m_0)

$m_0 = \left[\frac{(N_1+1)(n+1)}{(N+2)}\right]$ = Mode.

Limiting Condition of Hypergeometric Dist^n:-

$N \rightarrow \infty$ (very Large) N_1, P, x

$$\lim_{N\to\infty} \frac{\binom{NP}{x}\binom{Nq}{n-x}}{\binom{N}{n}}$$

$$\lim_{N\to\infty} \frac{NP(NP-1)\cdots(NP-x+1)\,Nq(Nq-1)\cdots(Nq-n+x+1)}{x!\times(n-x)!\times\frac{N(N-1)\cdots(N-n+1)}{n!}}$$

$$\frac{n!}{x!(n-x)!} \lim_{N\to\infty} \frac{\frac{NP}{NP}\left(1-\frac{1}{NP}\right)\cdots\left(1-\frac{x-1}{NP}\right)\frac{Nq}{Nq}\left(1-\frac{1}{Nq}\right)\cdots\left(1-\frac{n-x-1}{Nq}\right)}{\frac{N}{N}\left(1-\frac{1}{N}\right)\cdots\left(1-\frac{n-1}{N}\right)} \times p^x q^{n-x}$$

$\Rightarrow \binom{n}{x} p^x q^{n-x}$ (Binomial Distribution)

Problem:-

(1) During war 1 ship out of 9 sunk on an avg. what is the Prob. that exactly 3 out of a convoy of 6 ships would survive safely?

Soln:- $X = 3$.

$n = 6$

$p = 8/9$

$q = 1/9$

$P(X=3) = {}^6C_3 \times \left(\frac{8}{9}\right)^3 \left(\frac{1}{9}\right)^3$.

X = No. of ships that will survive in a convoy of 6 ships.

(Binomial Distribution)

---

(2) The Avg No. of misprints per Page of a book is 2. Assuming Poission Dist^n (i) what is the Prob that a particular Page is free from misprints if the book contains 500 Pages.

(ii) How many Pages contain more than 2 misprints

$\lambda = 2$

$n = 500$.

X = # misprints / Page.

$X \sim P(\lambda = 2)$

(i) $P(X=0) = \frac{e^{-2}\lambda^0}{0!} = e^{-2}$

(ii) $P_{(X>2)} = 1 - \left[P_{(X=0)} + P(X=1) + P(X=2)\right]$

$1 - \left[e^{-2} + \frac{e^{-2}(2)^1}{1!} + \frac{e^{-2}(2)^2}{2!} \right]$

(iii) Required no of Pages = $500 \times P(X>2)$.

59:30.

(3) Probablity of having a female child is $1/3^{rd}$ How many children are expected before the first female child?

X = no. of children before the first female child.

$X \sim Geo(P = 1/3)$

$E(X) = \frac{q}{p} = \frac{2 \times 3}{3 \times 1} = 2$.

if 3rd female child.

X = # children before 3rd female child

$X \sim Neg(r=3, P=\frac{1}{3})$

$E(X) = \frac{rq}{p} = \frac{2 \times 3}{1} = 6$.

Prob:- A person is paying Rs 10 for the following game. He is Drawing 2 cards from a deck

(i) He gets Rs. 20 if he draws 2 black queens
(ii) He gets Rs 30 if he draws 2 kings.
(iii) He gets Rs 40 if he draws 1 Red ace, 1 black ace.

what is his expected gain

X = gain.

$X = 10$ $(20-10)$
⇓ He drawn
$P = (2 \text{ black queen}) = \frac{1}{52C_2}$

$X = 20$ $P(2 \text{ kings}) = \frac{4C_2}{52C_2}$

$X = 30$ $P(1RA, 1BA) = \frac{2C_1 \times 2C_1}{52C_2}$

$$E(X) = 10 * \frac{1}{52C_2} + 20 \times \frac{4C_2}{52C_2} + 30 \times \frac{2C_1 \times 2C_1}{52C_2}$$

X	P(X=x)	X*P(X=x)	$X^2 P(X=x)$
10	$1/52C_2$		
20	$4C_2/52C_2$		
30	$\frac{2C_1 \times 2C_1}{52C_2}$		

$$Var(X) = E(X^2) - (E(X))^2$$

Qus.- A set of 8 coins are tossed The freq. dist^{n} of the no. of heads are given below. Fit the Binomial Distribution For this & comment

(x) Heads	Freq
0	1
1	5
2	25
3	65
4	66
5	48
6	35
7	8
8	2

$\hat{p} = \frac{1036}{255\,n} = \frac{1036}{8 \times 255} = 0.508$

$\hat{q} = 0.492$ $\quad n = 8$

X	f	fx	$\frac{n-x+1}{x}$	$1.032 \times$	$P(X=x)$	$N \times P(X=x)$
0	1	0	—	—	$(1-\hat{p})^n = 0.00343$	0.875
1	5	5	8	8.26	0.028	7.225
2	25	50	3.5	3.614	0.1012	25.80
3	65	195	2	2.064	0.209	53.26
4	66	264	1.25	1.291	0.269	68.75
5	48	240	0.8	0.8256	0.222	56.77
6	35	210	0.5	0.57625	0.115	29.30
7	8	56	2/7	0.295	0.034	8.645
8	2	16	1/8	0.129	0.0044	1.12
	255	1036			0.986	

$$\frac{P(X=x)}{P(X=x-1)} \Rightarrow \frac{{}^{n}C_{x} \cdot p^{x} q^{n-x}}{{}^{n}C_{x-1}\, p^{x-1} q^{n-x+1}}$$

$$= \frac{\frac{1}{x!(n-x)!}}{\frac{1}{(x-1)!(n-x+1)!}} \times \frac{p}{q} = \frac{n-x+1}{x} \times \frac{p}{q}$$

Qus:-

Fit Poission Distribution / negative Binomial Distribution to the following Defect Data.

x	Freq.	$\frac{\hat{\lambda}}{x}$	$P(x=x)$	$N \times P(x=x)$ (Exp Freq)
0	213	-	$e^{-\hat{\lambda}}$	
1	128			
2	37			
3	18			
4	3			
5	1			
>5	0			
			1	N.

Total

$$P(x=x) = \frac{\hat{\lambda}}{x} P(x=x-1)$$

$$\boxed{\bar{x} = \hat{\lambda}} = \frac{\Sigma fx}{\Sigma F}$$

NOTE

For ∞ Type of Data if Poission doesn't fit → Geometric → -ve Binomial

For Limited Data if Binomial doesn't fit → Hypergeometric.

Exercise:-

Draw the graph for the following Distribution

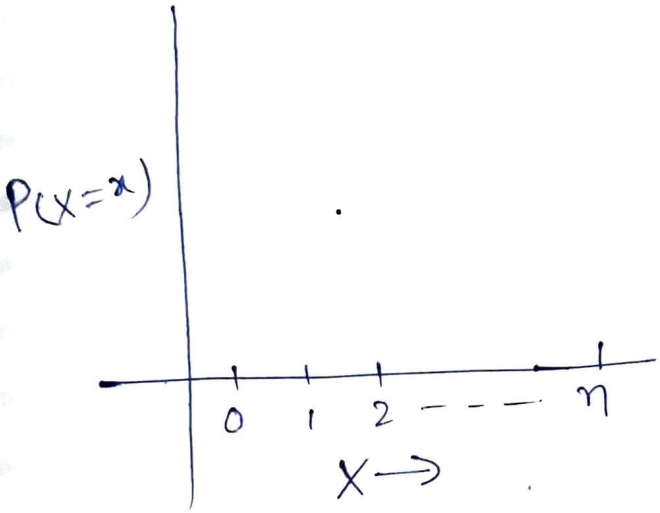
i) Bin ($n=10, P=0.5$)
ii) Bin ($n=10, P=0.2$)
iii) Bin ($n=10, P=0.9$)
iv) Pois ($\lambda=2$), Pois ($\lambda=10$)
v) Geo ($P=0.5$)
vi) Neg. Bin ($r=4, P=0.3$)

For $n \to$ large, $P \to$ small $\Big\} \to$ Pois ($\lambda = nP$)

- if $P=q$ is Bin $\to$ Sym.

$$\text{Skewness} = \frac{\mu_3^2}{\mu_2^3} = 0$$

$$\boxed{\mu_3 = 0}$$



Prob:- Two athlatic Teams A & B play a series of independent games until one of them wins 4 games. Prob of winning of each team is $\frac{1}{2}$. Find the Prob. that the series will end in at most 6 Games.

4G + 5G + 6G

$\left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + {}^4C_3 \times \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)$

Option-I

$+ {}^4C_3 \times \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)$

$+ {}^5C_3 \times \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)$

$+ {}^5C_3 \times \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)$

3A, 1B. A ⇒

3B, 1A. B

3A + 2B. A

Prob: A & B throw alternatively a pair of dice. A wins if he scores 6 points (Sum) before B gets 7 points. B wins if he gets 7 points before A gets 6 points. If A starts the game what is the Prob that he will win?

11	21	31	41	(51)	61
12	22	32	(42)	52	62
13	23	(33)	43	53	63
14	(24)	34	44	54	64
(15)	25	35	45	55	65
16	26	36	46	56	66

$P(A) = 5/36$ = getting 6.

$P(B) = 6/36$ = getting 7

$P(A) + P(\bar{A}\bar{B}A) + P(\bar{A}\bar{B}\bar{A}\bar{B}A) - - -$

$\frac{5}{36} + \frac{31}{36} \times \frac{30}{36} \times \frac{5}{36} + \left(\frac{31}{36} \times \frac{30}{36}\right)^2 \times \frac{5}{36} + - -$

$\Rightarrow \frac{5/36}{1 - \frac{31 \times 30}{36 \times 36}} = \frac{5/36}{\frac{36 \times 36 - 31 \times 30}{36 \times 36}}$

$\Rightarrow \frac{5}{1} \times \frac{36}{36^2 - 31 \times 30}$

$\Rightarrow \frac{5 \times 36}{366} \Rightarrow \left(\frac{30}{61}\right)$

Cummulative Distribution function (C d.f)

$$F(x) = P(X \leq x)$$

For discrete dist^n

$$F(x) = \sum_{x=n}^{x} P(x=x)$$

Univariate Continuous Distribution:-

X is having is c.d.f $F(x)$, it is said to be continuous R.V if F is continuous

$$f(x) = \frac{dF(x)}{dx} \rightarrow \text{Prob. density fu}^n.$$

$$f(x) = P(x-\Delta x \leq X \leq x+\Delta x)$$

X is called a cont. Random variable if it can have values within a range with sp. Prob. The con. dist^n fu^n $F(x)$ must be continuous.

Prop. of a PdF ($f(x)$):-

$$\int_{-\infty}^{\infty} f(x) \cdot dx = 1.$$

(I) Uniform or Rectangular Dist^n:-

X is said to follow uniform dist^n within the range (a,b) if

$$\left[f(x) = \frac{1}{b-a} \quad a \le x \le b \right]$$

$X \sim u(a,b)$ or $R(a,b)$

R-1 $f(x)$ is a pdf

$$\int_a^b \left(\frac{1}{b-a}\right) dx = 1$$

For discrete

$$E(X) = \sum x\, P(x=x)$$

For cont.

$$E(\varphi(x)) = \int_{-\infty}^{\infty} \varphi(x) f(x)\, dx$$

R-2

$E(X)$ for $u(a,b)$

$$E(X) = \int_a^b x \cdot \frac{1}{b-a}\, dx$$

$$\left(\frac{1}{b-a}\right)\left(\frac{x^2}{2}\right)_a^b$$

$$\boxed{E(X) = \frac{a+b}{2}}$$

R-3

$Var(X)$ for $u(a,b)$

$$Var(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \int_a^b x^2 \left(\frac{1}{b-a}\right) dx = \frac{(b^3-a^3)}{3(b-a)} = \frac{(b^2+a^2+ab)}{3}$$

$$Var(X) = \frac{a^2+b^2+ab}{3} - \left(\frac{a+b}{2}\right)^2$$

$$= \frac{a^2+b^2+ab}{3} - \left(\frac{a^2+b^2+2ab}{4}\right)$$

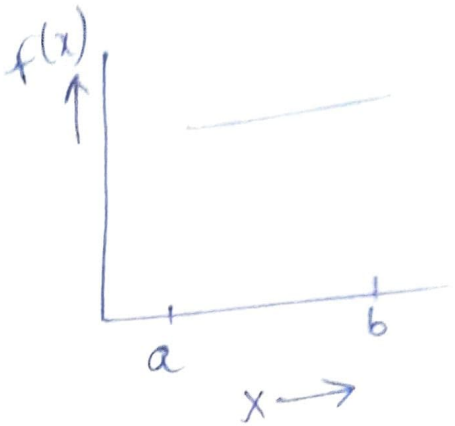
$$= \frac{4a^2+4b^2+4ab-3a^2-3b^2-6ab}{12}$$

$$= \frac{a^2+b^2-2ab}{12} = \frac{(b-a)^2}{12} = \frac{R^2}{12} \quad (R \Rightarrow \text{Range})$$

CDF

$$F(x) = \int_a^x \left(\frac{1}{b-a}\right) \cdot dx$$

$$= \frac{x-a}{b-a}$$



$$P(a_1 \le x \le b_1) = \frac{b_1-a_1}{b-a}$$

$$[a \le a_1 \le b_1 \le b] \quad = P(x \le b_1) - P(x \le a_1)$$

Skewness → Symmetric
Kurtosis → Platypuntik.

$$\text{Skewness} = \frac{\mu_3^2}{\mu_2^3}$$

$$\text{Kurtosis} = \frac{\mu_4}{\mu_2^2}$$

$$M_X(t) = E(e^{tx})$$

$$= \int_a^b e^{tx} \left(\frac{1}{b-a}\right) dx$$

$$= \left(\frac{1}{b-a}\right) \frac{e^{tx}\Big|_a^b}{t}$$

$$M_X(t) = \frac{1}{(b-a)t}\left[e^{bt} - e^{at}\right]$$

Find Measure of skewness & Kurtosis for uniform dist^n (Exercise).

For Continuous dist^n

$$E(\varphi(x)) = \int_{-\infty}^{\infty} \varphi(x) \cdot f(x) \cdot dx$$

$$\mu_r' = \int_{-\infty}^{\infty} x^r f(x) \cdot dx$$

$$\mu_r = \int_{-\infty}^{\infty} (x-\mu)^r f(x)\, dx$$

$$M_x(t) = \int_{-\infty}^{\infty} e^{tx} f(x)\, dx$$

$$\text{Median}(\tilde{\mu}) \Rightarrow \int_{-\infty}^{\tilde{\mu}} f(x) \cdot dx = \frac{1}{2}$$

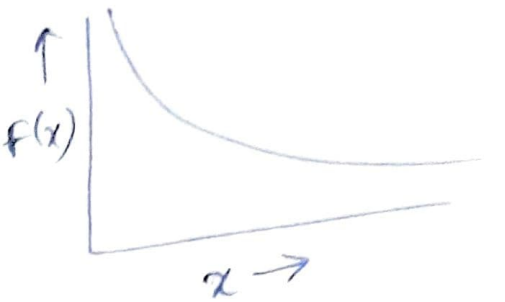
$$\text{Mode}(\breve{\mu}) \Rightarrow \frac{d}{dx} f(x) = 0 \Rightarrow x \checkmark \quad \text{(check Second diff} \rightarrow -ve)$$

$$\max_x f(x) \checkmark$$

(11) Exponential Distribution:-

X is said to follow exponential dist^n with a Parameter λ

$$f(x) = \lambda e^{-\lambda x} \quad 0 < x < \infty$$



R-1 $f(x)$ is a p.d.f

$$\int_0^\infty f(x)\,dx = \int_0^\infty \lambda e^{-\lambda x}\,dx$$

$$= \frac{\lambda}{-\lambda}\left(e^{-\lambda x}\right)_0^\infty$$

$$= 1.$$

Mean & Var of exp(λ):-

MGF of exp(λ)

$$M_X(t) = E(e^{tx})$$

$$= \int_0^\infty e^{tx} \cdot \lambda \cdot e^{-\lambda x}\,dx$$

$$= \lambda \int_0^\infty e^{-x(\lambda - t)}\,dx$$

$z = x(\lambda - t)$

$\frac{dz}{dx} = (\lambda - t)$

$$= \lambda \int_0^\infty e^{-z} \frac{dz}{(\lambda - t)}$$

$$= \left(\frac{\lambda}{\lambda - t}\right) \int_0^\infty e^{-z}\,dz$$

$$= \frac{\lambda}{\lambda - t} = \frac{1}{1 - \frac{t}{\lambda}} = 1 + \frac{t}{\lambda} + \frac{t^2}{\lambda^2} + \cdots$$

$$M_X(t) = \sum_{r=0}^{\infty} \frac{t^r}{r!} \mu_r'$$

$$\mu_r' = \frac{r!}{\lambda^r}$$

$$\mu_1' = \frac{1}{\lambda} = E(X)$$

$$\mu_2' = \frac{2}{\lambda^2}$$

$$Var(X) = \mu_2$$

$$= \mu_2' - (\mu_1')^2$$

$$\boxed{Var(X) = \frac{1}{\lambda^2}}$$

c.d.f of $exp(\lambda)$

$$P = F(x) = P(X \le x) = \int_0^x \lambda \cdot e^{-\lambda x} \, dx$$

$$= \lambda \frac{e^{-\lambda x}}{-\lambda} \Big|_0^x$$

$$= 1 - e^{-\lambda x}$$

$$P(X \le a) = 1 - e^{-\lambda a}$$

$$P(a \le X \le b) = P(X \le b) - P(X \le a)$$

$$= e^{-\lambda a} - e^{-\lambda b}$$

Memory Less Property of exponential dist^n:-

Let X Represents the life dist^n of a Component.

Prob. that the Component will Survive for atleast another b units given that it has already Survived for a units is same as the Prob. that it will Survive for atleast b units starting from 0.

$$\boxed{P(X \ge a+b / X \ge a) = P(X \ge b)}$$

Proof $P(x \le a) = 1 - e^{-\lambda a}$

$P(x \ge a) = e^{-\lambda a}$

L.H.S. $\Rightarrow \dfrac{P(x \ge a+b)}{P(x \ge a)}$

$= \dfrac{e^{-\lambda(a+b)}}{e^{-\lambda a}} = e^{-\lambda b} = P(x \ge b)$

For cont dist^n:-

$P(a \le x \le b) = P(a < x \le b) = P(a \le x < b) = P(a < x < b)$ all same

How to fit a data on exp$^{(n)}$

X	freq.	$P(a_i \le x \le a_i')$	exp fr-eq $= N \times P(a_i \le x \le a_i')$
~~0 – a_1~~	f_0	$1 - e^{-\lambda a_1}$	
$a_1 - a_2$	f_1	$e^{-\lambda a_1} - e^{-\lambda a_2}$	
$a_2 - a_3$	f_2	⋮	
$a_3 - a_4$	f_3		
$a_4 - a_5$	f_4		
$> a_5$	0	(Adjust here)	
	N	1	

$X \sim exp(\lambda)$

$E(X) = \dfrac{1}{\lambda}$

$P(a \le x \le b) = e^{-\lambda a} - e^{-\lambda b}$.

$\bar{x} = \dfrac{\sum x_i f_i}{\sum f_i} = \dfrac{1}{\hat{\lambda}}$ $\quad \boxed{\hat{\lambda} = \dfrac{1}{\bar{x}}}$

Then Compare freq., exp freq. variation.

Gamma Distribution:-

(III)

Gamma fun $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} \, dx$

$= (n-1)!$

$= (n-1)\,\Gamma(n-1)$

X is said to follow Gamma dist^n with parameter α & P.

$$f(x) = \frac{\alpha^P}{\Gamma(P)} e^{-\alpha x} x^{P-1} \qquad 0 < x < \infty.$$

R-1 $f(x)$ is a Pdf

$$\int_0^\infty f(x)\,dx = \frac{\alpha^P}{\Gamma(P)} \int_0^\infty e^{-\alpha x} x^{P-1} \, dx.$$

$\alpha x = z$, $\alpha\,dx = dz$

$$= \frac{\alpha^P}{\Gamma(P)} \int_0^\infty \frac{e^{-z} z^{P-1}}{\alpha^{P-1}} \cdot \frac{dz}{\alpha}$$

$$= \frac{1}{\Gamma(P)} \int_0^\infty e^{-z} z^{P-1} \, dz$$

$$= \frac{\Gamma(P)}{\Gamma(P)} = 1$$

Mean & Variance

$$E(X) = \int_{-\infty}^{\infty}$$

$$\mu_r' = \frac{\alpha^P}{\Gamma(P)} \int_0^\infty x^r e^{-\alpha x} x^{P-1} \, dx$$

$\alpha x = z$ $\quad \alpha\,dx = dz$

$$\mu_r' = \frac{\alpha^P}{\Gamma(P)} \int_0^\infty \left(\frac{z}{\alpha}\right)^{P+r-1} \cdot e^{-z} \frac{dz}{\alpha}$$

$$= \frac{1}{\Gamma(P)\,\alpha^r} \int_0^\infty e^{-z} z^{P+r-1} \, dz.$$

$$\mu_r' = \frac{\Gamma(p+r)}{\Gamma(p)\cdot \alpha^r}$$

Mean $\Rightarrow \mu_1' = \left(\frac{p}{\alpha}\right)$

$$\mu_2' = \frac{p(p+1)}{\alpha^2}$$

$Var(x) = \frac{p}{\alpha^2}$

$$Var(x) = \mu_2' - (\mu_1')^2$$

$$= \frac{p^2(p+1)^2}{\alpha^2} - \frac{p^2}{\alpha^2}$$

$$= \frac{p^2+p-p^2}{\alpha^2} = \frac{p}{\alpha^2}$$

Estimate the Parameters of Gamma dist^n from sample data Find $\bar{x}, s^2$

$$\frac{\hat{p}}{\hat{\alpha}} = \bar{x}, \quad \frac{\hat{p}}{\hat{\alpha}^2} = s^2 \Rightarrow \hat{p}, \hat{\alpha}$$ ✓

MGF of Gamma Distribution:-

$X \sim Gamma(\alpha, p)$

$$M_X(t) = \int_0^\infty e^{tx} \frac{\alpha^p}{\Gamma(p)} e^{-\alpha x} \cdot x^{p-1} dx$$

$$= \frac{\alpha^p}{\Gamma(p)} \int_0^\infty e^{-x(\alpha-t)} \cdot x^{p-1} \cdot dx$$

$z = x(\alpha-t)$

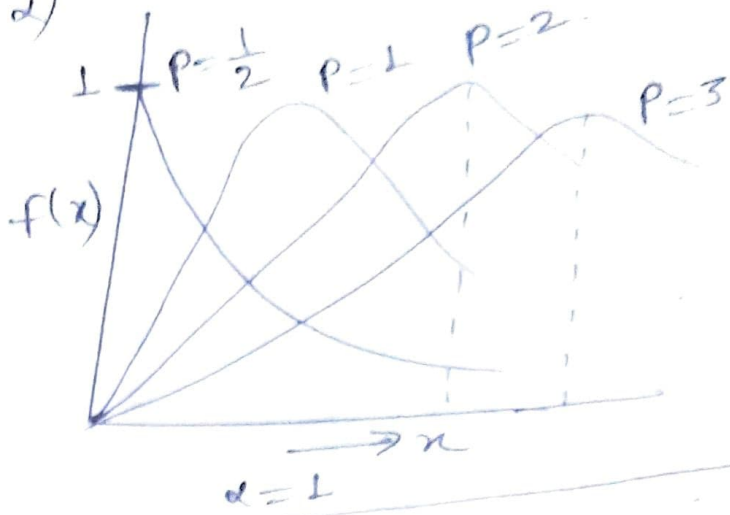
$\frac{dz}{dx} = (\alpha-t)$

$$= \frac{\alpha^p}{\Gamma(p)} \int_0^\infty \frac{e^{-z} \cdot z^{p-1}}{(\alpha-t)^{p-1}} \cdot \frac{dz}{(\alpha-t)}$$

$$= \frac{\alpha^p}{\Gamma(p)(\alpha-t)^p} \int_0^\infty e^{-z} \cdot z^{p-1}\, dz$$

$$= \frac{\alpha^p}{(\alpha-t)^p} = \frac{1}{\left(1-\frac{t}{\alpha}\right)^p} = \left(1-\frac{t}{\alpha}\right)^{-p}$$

$$\boxed{M_x(t) = \left(1-\frac{t}{\alpha}\right)^{-p}}$$



(IV) Beta Distribution:-

$$B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx$$

$$B_1(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx \quad \text{(First kind)}$$

$$B_2(m,n) = B(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\, dx \quad \text{(Second kind)}$$

X is said to follow Beta dist^n of 1st kind if

$$\boxed{f(x) = \frac{1}{B(m,n)} x^{m-1}(1-x)^{n-1}} \quad 0<x<1$$

Mean & variance

r^{th} order Raw Moment $\mu_r' = E(x^r) = \int_0^1 \frac{x^r\, x^{m-1}(1-x)^{n-1}}{B(m,n)}\, dx$

$$= \frac{1}{B(m,n)} \int_0^1 x^{m+r-1} (1-x)^{n-1} \, dx$$

$$= \frac{B(m+r,n)}{B(m,n)}$$

$$\mu_1' = \frac{B(m+1,n)}{B(m,n)} = \frac{\Gamma(m+1)\,\Gamma n}{\Gamma m+1+n \times \Gamma m\, \Gamma n} \cdot \frac{\Gamma m+n}{\Gamma m+n}$$

$$\boxed{\text{Mean} = \frac{m}{m+n}}$$

$$\mu_2' = \frac{B(m+2,n)}{B(m,n)} = \frac{\Gamma m+2\, \Gamma n\, \Gamma m+n}{\Gamma(m+n+2)\, \Gamma m\, \Gamma n}$$

$$= \frac{(m+1)m}{(m+n+1)(m+n)}$$

$$\left[\mu_2 = \mu_2' - (\mu_1')^2 = \text{var}(x)\right]$$

Beta dist^n of second kind

X is said to follow Beta dist^n of second kind

if $f(x) = \frac{1}{B(m,n)} \frac{x^{m-1}}{(1+x)^{m+n}}$ $\quad 0 < x < \infty$.

$$\mu_r' = \frac{1}{B(m,n)} \int_0^\infty \frac{x^{m+r-1}}{(1+x)^{m+n}} \cdot dx.$$

$$\mu_r' = \frac{B(m+r, n-r)}{B(m,n)}$$

11-Nov

Normal Distribution

X is R.V said to follow Normal dist^n with Parameter μ & σ if P.d.f

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \qquad -\infty < x < \infty$$

R-1 $f(x)$ is a P.d.f

Proof $\int_{-\infty}^{\infty} f(x).dx = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$

$z = \frac{x-\mu}{\sigma}$ (z-Transformation)

$dz = \frac{dx}{\sigma}$

$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz$

$= \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-z^2/2} dz$ (Even function) ($f(-x) = f(x)$)

$\int_{-\infty}^{\infty} f(x).dx = \frac{1}{\sqrt{2\pi}} (2) \int_{0}^{\infty} e^{-y} \frac{dy}{\sqrt{2y}}$

$y = z^2/2$

$= \frac{1}{\sqrt{\pi}} \int_{0}^{\infty} e^{-y} . y^{-1/2} . dy$

$\frac{dy}{dz} = z$

$dz = \frac{dy}{z} = \frac{dy}{\sqrt{2y}}$

$\Gamma\frac{1}{2} = \sqrt{\pi}$

$= \frac{\sqrt{\pi}}{\sqrt{\pi}} = 1.$

#Mean of $N(\mu, \sigma)$

$$E(x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} x . e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} . dx$$

$$Z = \frac{x-\mu}{\sigma}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + z\sigma) e^{-z^2/2} \, dz$$

$$= \frac{1}{\sqrt{2\pi}} \mu \int_{-\infty}^{\infty} e^{-z^2/2} \, dz + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z \, e^{-z^2/2} \, dz$$

odd fun^n $\Rightarrow 0$

$$= \frac{1}{\sqrt{2\pi}} \mu \int_{-\infty}^{\infty} e^{-z^2/2} \, dz$$

$$= \frac{2\mu}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-y} \frac{dy}{\sqrt{2y}}$$

$z^2/2 = y$

$z \, dz = dy$

$dz = \frac{dy}{z} = \frac{dy}{\sqrt{2y}}$

$$= \frac{\mu}{\sqrt{\pi}} \int_{0}^{\infty} e^{-y} \cdot y^{-1/2} \, dy$$

$\sqrt{\pi}$

$$= \mu.$$

Var of $N(\mu, \sigma^2)$

$$Var(X) = E(X-\mu)^2$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (x-\mu)^2 e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (z\sigma)^2 e^{-\frac{1}{2}(z)^2} \, dz.$$

$z = \frac{x-\mu}{\sigma}$

$$= \frac{\sigma^2}{\sqrt{2\pi}} (2) \int_{0}^{\infty} z^2 \cdot e^{-z^2/2} \, dz.$$

$$Var(X) = \frac{\sigma^2}{\sqrt{\pi}} 2 \int_{0}^{\infty} y \cdot e^{-y} \frac{dy}{\sqrt{2y}}$$

$y = z^2/2$

$dy = z \, dz$

$dz = \frac{dy}{\sqrt{2y}}$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^\infty e^{-y} \cdot y^{\frac{3}{2}-1} \, dy$$

$\Gamma\frac{3}{2}$

$$\frac{2\sigma^2}{\sqrt{\pi}} \times \frac{1}{2}\sqrt{\pi}$$

$$= \frac{2\sigma^2}{2} \Rightarrow \sigma^2 \quad (\text{var})$$

μ, σ^2 can be estimated by Sample Mean & Sample variance.

Odd ordered central Moment :-

$$\mu_{2r+1} = E(x-\mu)^{2r+1}$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (x-\mu)^{2r+1} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \cdot dx$$

$z = \frac{x-\mu}{\sigma}$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (z\sigma)^{2r+1} e^{-z^2/2} \cdot dz$$

Odd fun^n

$$= 0$$

$\boxed{\mu_3 = 0}$ $\left[\text{Skewness} = \frac{\mu_3^2}{\mu_2^3} = 0\right]$

$\Rightarrow$ Symmetric Curve

Even ordered central Moment

$$\mu_{2r} = E(x-\mu)^{2r}$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} (x-\mu)^{2r} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \cdot dx$$

$z = \frac{x-\mu}{\sigma}$

$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (z\sigma)^{2r} e^{-z^2/2} \, dz$

$= \frac{\sigma^{2r}}{\sqrt{2\pi}} (2) \int_{0}^{\infty} z^{2r} e^{-z^2/2} \, dz$

$\frac{z^2}{2} = y$

$dz = \frac{dy}{\sqrt{2y}}$

$= \frac{\sigma^{2r}}{\sqrt{2\pi}} (2) \int_{0}^{\infty} (2y)^r e^{-y} \frac{dy}{\sqrt{2y}}$

$= \frac{(\sigma^{2r})(2)^r}{\sqrt{\pi}} \int_{0}^{\infty} e^{-y} \cdot y^{r-\frac{1}{2}} \, dy.$

$= \frac{2^r \sigma^{2r}}{\sqrt{\pi}} \boxed{\left(r+\frac{1}{2}\right)}$

$= \frac{2^r \sigma^{2r}}{\sqrt{\pi}} \left(r-\frac{1}{2}\right)\left(r-\frac{3}{2}\right) \cdots \frac{1}{2} \sqrt{\pi}$

$= \frac{2^r \sigma^{2r} (2r-1)(2r-3) \cdots 1}{2^r}$

$= (2r-1)(2r-3) \cdots 1 \cdot \sigma^{2r}.$

$\mu_4 = 3\sigma^4$

Normal dist^n is a (Mesokurtic) dist^n.

kurtosis $\Rightarrow \frac{\mu_4}{\mu_2^2} = \frac{3\sigma^4}{\sigma^4} = 3.$

MGF of $N(\mu, \sigma^2)$

$$M_X(t) = E(e^{tX})$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \cdot dx$$

Z-Transformation. $\frac{x-\mu}{\sigma} = z$.

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(z\sigma+\mu)} \cdot e^{-z^2/2} (dz)$$

$$= \frac{e^{t\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t z\sigma - \frac{z^2}{2}} dz$$

$$= \frac{e^{t\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z^2 - 2tz\sigma)} \cdot dz$$

$$= \frac{e^{t\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left((z-t\sigma)^2 - t^2\sigma^2\right)} \cdot dz$$

$$= \frac{e^{t\mu + \frac{t^2\sigma^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-t\sigma)^2} \cdot dz$$

$y = z - t\sigma$

$$= \frac{e^{t\mu + \frac{t^2\sigma^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-y^2/2} dy.$$

$y' = y^2/2$

$$\boxed{M_X(t) = e^{t\mu + \frac{t^2\sigma^2}{2}}}$$

$\mu_r' = ?$

MGF About μ

$$M_{X-\mu}(t) = E\left(e^{t(x-\mu)}\right)$$

$$= \int_{-\infty}^{\infty} e^{t(x-\mu)} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\frac{e^{-t\mu}}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{tx} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= e^{-t\mu} M_X(t)$$

$$= e^{-t\mu} e^{t\mu + \frac{t^2\sigma^2}{2}}$$

$$= e^{\frac{t^2\sigma^2}{2}}$$

$$\Rightarrow \sum_{0}^{\infty} \frac{t^r}{r!} \mu_r \quad \text{(By definition)}$$

$$\Rightarrow \sum_{0}^{\infty} \frac{t^r}{r!} \mu_r'$$

$$M_X(t) = \sum_{0}^{\infty} \frac{t^r}{r!} \mu_r'$$

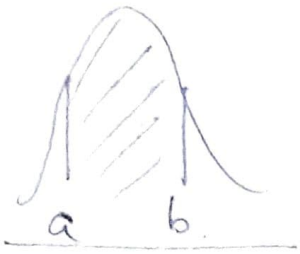
$$M_{X-\mu}(t) = \sum_{0}^{\infty} \frac{t^r}{r!} \mu_r$$

$$\mu_r' = \frac{d^r}{dt^r} M_X(t)\Big|_{t=0}$$

$$\mu_r = \frac{d^r}{dt^r} M_{X-\mu}(t)\Big|_{t=0}$$

$X \sim N(\mu, \sigma^2)$ ⇒ Spec. Limit (LSL, USL)

$P(a \le X \le b)$ interested to find out percentage Non conformance for the quality characteristic.



$\Rightarrow 1 - P(LSL \le X \le USL)$

$$P(a \le x \le b) = \int_a^b \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \cdot dx$$

$$z = \frac{x-\mu}{\sigma} = \int_{\frac{a-\mu}{\sigma}}^{\frac{b-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \, dz$$

⇒ For $N(0,1)$

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \qquad -\infty < z < \infty.$$

$$\int_{a^*}^{b^*} \phi(z) \cdot dz.$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ $\quad a^* = \frac{a-\mu}{\sigma}$ $\quad b^* = \frac{b-\mu}{\sigma}$

$$P(a \le x \le b) = P\left(\frac{a-\mu}{\sigma} \le \frac{x-\mu}{\sigma} \le \frac{b-\mu}{\sigma}\right)$$

$$= \int_{\frac{a-\mu}{\sigma}}^{\frac{b-\mu}{\sigma}} \phi(z) \cdot dz$$

$\frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the pdf of std Normal or $z \sim N(0,1)$

$$= \int_{a^*}^{b^*} \phi(z)\,dz$$

$\phi \rightarrow$ pdf of $N(0,1)$

$$= P(z \le b^*) - P(z \le a^*)$$

$$= \int_{-\infty}^{b^*} \phi(z)\,dz - \int_{-\infty}^{a^*} \phi(z)\,dz$$

$$\Rightarrow \Phi(z) = \int_{-\infty}^{z} \phi(z)\,dz$$

$$= \Phi(b^*) - \Phi(a^*)$$

Standard Normal

Z is said to follow std Normal dist^n if

$Z \sim N(0,1)$ $\quad \phi(z) =$ pdf of Z.

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \quad [-\infty < z < \infty]$$

$\Phi(z) \Rightarrow$ cdf of Z.

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz.$$

Problem $\rightarrow$ Z is a std Normal

$$P(a \le z \le b)$$

$$= \Phi(b) - \Phi(a)$$

$$= \int_a^b \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz = \int_{-\infty}^{b} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz - \int_{-\infty}^{a} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz$$

$$= \Phi(b) - \Phi(a)$$

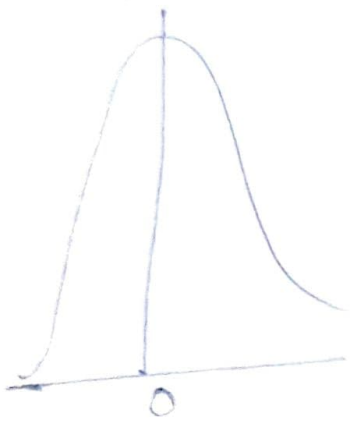
$X \sim N(\mu, \sigma^2)$

$$P(a \le X \le b) = P\left(\frac{a-\mu}{\sigma} \le \frac{X-\mu}{\sigma} \le \frac{b-\mu}{\sigma}\right)$$

$$\int_a^b \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = P(a^* \le z \le b^*)$$

$$= \Phi(b^*) - \Phi(a^*)$$

$P(z \le 0) = \frac{1}{2} = \Phi(0)$



$\Phi(1.75) = 0.9599$

$P(0.12 \le z \le 1.34) \Rightarrow \Phi(1.34) - \Phi(0.12)$

$= 0.9099 - 0.5478$

0.5478

$= 0.3621$

Median of Normal distribution $\Rightarrow \mu$ = Mean

For std Normal distribution Median = 0.

$$z = \frac{x-\mu}{\sigma}$$

$$x = \mu + z\sigma$$

[Median = μ]

[$\Phi(-1.21) = 1 - \Phi(1.21)$]

In Some Tables

$\Phi(0) = 0$

Here instead of original Φ, they are calculating integration from 0 to z.

Actual Φ is $\Rightarrow \Phi(z) + 0.5$.

From the table we can say Median of $N(0,1)$ is 0.

Since $\Phi(0) = 0.5$

$$z = \frac{x - \mu}{\sigma}$$

Median for $X \Rightarrow \mu + z\sigma$

$\Rightarrow \mu$

Exercise - Find out Mode for $N(\mu, \sigma^2)$

diff & equate to zero.

$$\boxed{\text{Mode} = \mu}$$

$$\boxed{\text{Mean} = \text{Median} = \text{Mode}}$$

Prob:-

Let x be a quality characteristic which follows normal distribution $\mu = 10, \sigma = 3$

Spec limit is 7 to 12 what is the Proportion non conformance.

$$P(7 \le x \le 12)$$

$$P\left(\frac{7-10}{3} \le \frac{x-\mu}{\sigma} \le \frac{12-10}{3}\right)$$

$$P\left(-1 \le \frac{x-\mu}{\sigma} \le \frac{2}{3}\right)$$

$$P(-1 \le \frac{x-\mu}{\sigma} \le 0.67)$$

$$\Phi(0.67) - \Phi(-1)$$

$$0.7486 - 0.1587$$

$$- \underline{0.1587}$$

$$0.5899 \Rightarrow \text{Proportion of conformance}$$

Prop. of Non. conf $\Rightarrow 1 - 0.5899$.

22/11/2021

Normal Distribution:-

A Standard Normal Distribution is a Normal distribution with mean=0, $\sigma = 1$.

Notation $Z \sim N(0,1)$

Convertion of $N(\mu, \sigma^2) \rightarrow N(0,1)$

$$Z = \frac{X-\mu}{\sigma}$$ where Z follow $N(0,1)$

$X \sim N(\mu, \sigma^2)$

For Std Normal PdF is $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ $-\infty < z < \infty$.

cdf is $\Phi(z) = \int_{-\infty}^{\infty} \phi(z) \cdot dz$.

$$P(a \le x \le b) = P\left(\frac{a-\mu}{\sigma} \le \frac{x-\mu}{\sigma} \le \frac{b-\mu}{\sigma}\right)$$

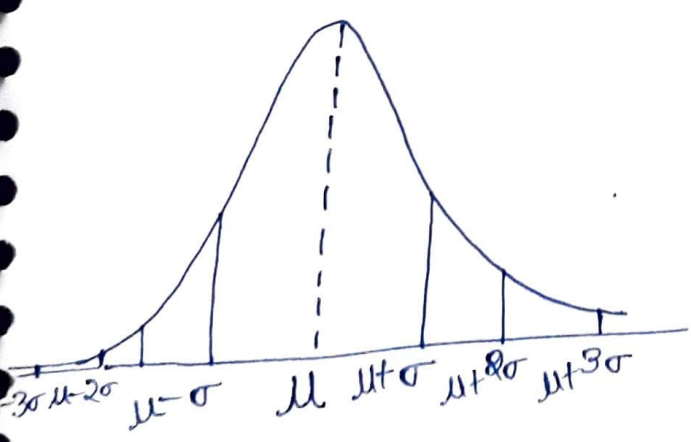
$$= P(a^* \le z \le b^*)$$

$$a^* = \frac{a-\mu}{\sigma}$$

$$b^* = \frac{b-\mu}{\sigma}$$

$$= \Phi(b^*) - \Phi(a^*)$$

Tabulated value of $\Phi(z)$ for different values of z are given in the Normal Table.



$X \sim N(\mu, \sigma^2)$

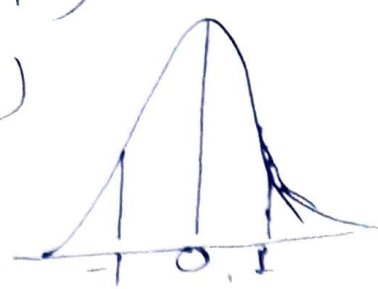
$P(\mu - \sigma \le X \le \mu + \sigma)$

$= P(-1 \le z \le 1)$

$= \Phi(1) - \Phi(-1)$

$= 0.8413 - 0.1587$

$= 0.6826$



$P[(\mu-2\sigma) \le x (\mu+2\sigma)] = 0.9545$.

$P(\mu-3\sigma \le X \le \mu+3\sigma) = 0.9973$. $= 99.73\%$. $= \Phi(3) - \Phi(-3)$

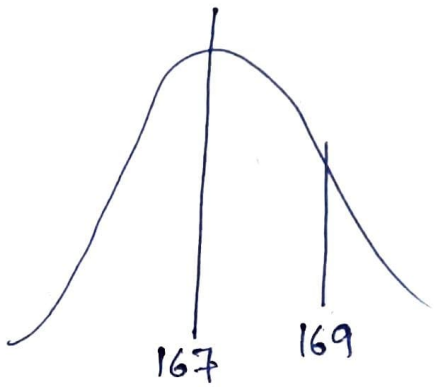
$\Rightarrow$ ~~99.3~~ 99.73% observation Lie within $\mu \pm 3\sigma$ for $N(\mu, \sigma^2)$.

Prob:-

(1) The height of men for a particular class follows normal ($\mu = 167$ cm, Std Dev = 3 cm)
what is the Proportion of the men whose height

(i) greater than 169 cm

(ii) in between (167 - 170) cm.



(i) ≥ 169 cm

$z = \frac{169-167}{\sigma} = \frac{2}{3}$

$z = 0.666$.

$= 0.67$.

$= 0.7486$.

Required Proportion $= 1 - 0.7486 = 0.2514$

$= 25.14\%$.

(ii) $P(167 \le X \le 170)$

$\Rightarrow P\left(\frac{167-167}{3} \le z \le \frac{170-167}{3}\right)$

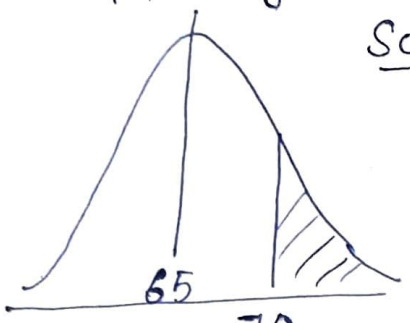
$\Rightarrow P(0 \le z \le 1)$

$\Rightarrow$ ~~$P(1) - P(0)$~~ $\Phi(1) - \Phi(0)$

$\Rightarrow 0.8413 - 0.50 = 0.3413$.

$= 34.13\%$

Qus(2) The Score distribution of certain class follows Normal ($\mu = 65, \sigma = 5$) if 3 students are selected at random. what is the probability that 2 out of them get more than 70?



Solution:- $X \rightarrow$ score

$$P(\text{Marks} \geq 70) = 1 - P(X \leq 70)$$

$$= 1 - P\left(Z \leq \frac{70-65}{5}\right)$$

$$= 1 - \Phi(1).$$

$$p = 1 - 0.8413.$$

$$p = 0.1587.$$

Req^{d} Prob :- $\binom{3}{2} p^2 (1-p)^1$.

Prob-(3) Suppose the duration (in Min) of a Telephone call $\sim exp(\lambda = \frac{1}{5})$ Find the Probability that the duration will exceed 5 min.

$X \sim$ Duration in Min.

$X \sim exp(\lambda = 1/5)$

$$P(X > 5) = \int_5^\infty \lambda e^{-\lambda x} \, dx.$$

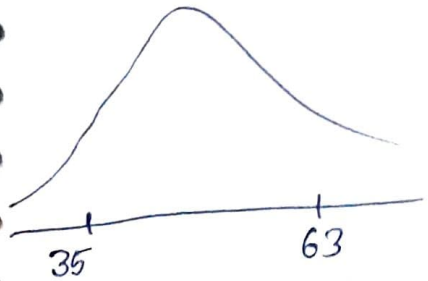
Qus(4) A Quality characteristics (x) under study follows $N(\mu, \sigma^2)$

Specification limit : 35-63

7% falls below Low Spec Limit.

11% falls above USL.

Find μ & σ.



$X \sim N(\mu, \sigma^2)$

$P(X \leq 35) = 0.07$

$\Phi\left(\frac{35-\mu}{\sigma}\right) = 0.07$

$\frac{35-\mu}{\sigma} = -1.48$

$P(X \geq 63) = 0.11$

$P(X \leq 63) = 0.89$

$P\left(\frac{X-\mu}{\sigma} \leq \frac{63-\mu}{\sigma}\right) = 0.89.$

$\frac{63-\mu}{\sigma} = 1.23$

$(\mu = 50.29)$

Fitting Normal Distribution:-

X	obs Freq.	x_i	$f_i x_i$	$p(a \le x \le b)$
~~<4.5~~	0			
4.5 – 9.5	1	7	7	
9.5 – 14.5	3	12	36	
14.5 – 19.5	24	17	408	
19.5 – 24.5	58	22	1276	
24.5 – 29.5	60	27	1620	
29.5 – 34.5	27	32	864	
34.5 – 39.5	2	37	74	
39.5 – 44.5	2	42	84	
~~>44.5~~	0			
	$177 = N$			

Find out $\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = 24.68$. $\quad S = 5.\text{--}$

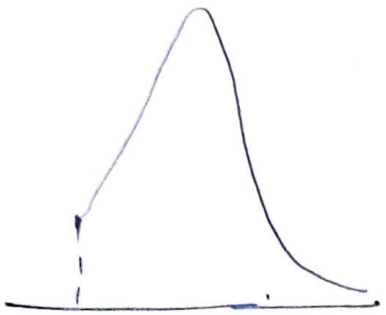
$$S^2 = \frac{1}{N-1} \sum_i (x_i - \bar{x})^2 f_i$$

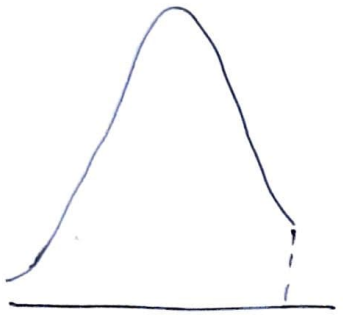
[S^2 is an unbaised estimate of σ^2]

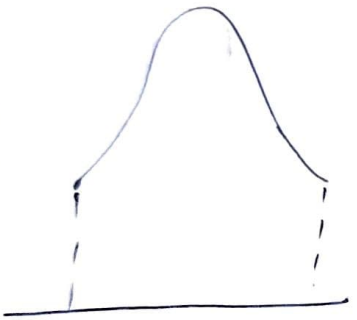
X	$Z = \frac{X - \bar{X}}{S}$	$\Phi(z)$	$\nabla \Phi(z)$ Exp Rel. Freq	~~$p(a \le x \le b)$~~	$N \times \nabla \Phi(z)$ Exp Freq.
$-\infty$	$-\infty$	0			
4.5	$\frac{4.5 - \bar{x}}{s}$	–			
9.5	⋮	–			
14.5	⋮	–			
19.5	⋮	–			
24.5	⋮	–			
29.5	⋮				
34.5	⋮				
39.5	⋮				
⋮	⋮				
44.5	⋮				
∞		1	1		N

Now Compare obs. Freq. & Exp. Frequency variation.
Varidion.

Truncated Distribution:-







For Discrete case.

X is said to follow Binomial Distribution with parameters n & p truncated to the left of $X = c$. if the values for man points $X = 0, 1, 2 \ldots (c-1)$ are excluded from the Distribution.

P.M.F $\quad g_1(x) = K_1 \binom{n}{x} p^x q^{n-x}$

$x = c, c+1 \ldots n.$

$$\sum_x g_1(x) = 1$$

$$\sum_{c}^{n} K_1 \binom{n}{x} p^x q^{n-x} = 1.$$

$$K_1 = \frac{1}{\sum_{c}^{n} \binom{n}{x} p^x q^{n-x}}$$

X is said to follow Bin. Dist. (n,p) truncated to the right of $X = d$, then values for man points $X = d+1, d+2 \ldots n$ are excluded from the distribution.

$$g_2(x) = K_2 \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, \ldots d.$$

considering prop. of pmf

$$\sum g_2(x) = 1$$

$$K_2 \sum_{x=0}^{d} \binom{n}{x} p^x q^{n-x} = 1.$$

$$K_2 = \frac{1}{\sum_{0}^{d} \binom{n}{x} p^x q^{n-x}}.$$

X is said to follow Binomial Dist^n with Parameters n & P. Truncated to the left of $x=c$ & to the right of $x=d$ then only $x = c, c+1, \ldots d$ points are included.

$$g_3(x) = K_3 \binom{n}{x} p^x q^{n-x} \qquad x = c, c+1, \ldots d.$$

$$K_3 \sum_{c}^{d} \binom{n}{x} p^x q^{n-x} = 1$$

$$\boxed{K_3 = \frac{1}{\sum_{c}^{d} \binom{n}{x} p^x q^{n-x}}}$$

Truncated Normal Distribution:-

X is said to follow a normal dist^n with Parameters μ & σ^2, truncated to the

(i) left of $x=c$, if

$$f_1(x) = K_1 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$c \le x \le \infty.$$

Following Prop. of P.M.F

$$\int_c^\infty f_1(x)\,dx = 1 \Rightarrow K_1 \int_c^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 1.$$

$$K_1 = \frac{1}{1-\Phi(c^*)} \qquad \left(c^* = \frac{c-\mu}{\sigma}\right)$$

(ii) Right of $x=d$ if

$$g_2(x) = K_2 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \qquad [-\infty \le x \le d]$$

$$K_2 \int_{-\infty}^{d} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 1. \Rightarrow \boxed{K_2 = \frac{1}{\Phi(d^*)}} \qquad \left(d^* = \frac{d-\mu}{\sigma}\right)$$

(iii) Left of $x=c$ & Right of $x=d$

new pdf $g_3(x) = K_3 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad c \le x \le d.$

$$\int_c^d g_3(x)\,dx = 1.$$

$$K_3 \int_c^d \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \cdot dx = 1.$$

or $\boxed{K_3 = \frac{1}{\Phi(d^*) - \Phi(c^*)}}$ $\rightarrow d^* = \frac{d-\mu}{\sigma}$ $\rightarrow c^* = \frac{c-\mu}{\sigma}$

⊕ Expectation of Both Side Truncated Normal Dist^n:-

$$\Rightarrow \int_c^d x \cdot K_3 \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \qquad K_3 = \frac{1}{\Phi(d^*) - \Phi(c^*)}$$

Take $z = \frac{x-\mu}{\sigma}$

$$\Rightarrow K_3 \int_{\frac{c-\mu}{\sigma}}^{\frac{d-\mu}{\sigma}} (\mu + z\sigma) \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \cdot dz.$$

$$\Rightarrow K_3 \frac{\mu}{\sqrt{2\pi}} \int_{c^*}^{d^*} e^{-z^2/2} \cdot dz + \frac{\sigma K_3}{\sqrt{2\pi}} \int_{c^*}^{d^*} z \cdot e^{-z^2/2}\, dz.$$

$$\Rightarrow \mu + \frac{\sigma K_3}{\sqrt{2\pi}} \int_{c^*}^{d^*} z \cdot e^{-z^2/2} \cdot dz. \qquad Y = \frac{z^2}{2}$$

$$\Rightarrow \mu + \frac{\sigma K_3}{\sqrt{2\pi}} \int_{\frac{c^{*2}}{2}}^{\frac{d^{*2}}{2}} e^{-y}\, dy. \qquad dy = dz.(\bullet z)$$

$$\Rightarrow \mu + \frac{\sigma K_3}{\sqrt{2\pi}} \left(e^{-y}\right)_{\frac{c^{*2}}{2}}^{\frac{d^{*2}}{2}} \Rightarrow \mu + \frac{\sigma}{\Phi(d^*) - \Phi(c^*)} \left(\frac{1}{\sqrt{2\pi}} e^{-c^{*2}/2} - \frac{1}{\sqrt{2\pi}} e^{-d^{*2}/2} \right)$$

$$E \Rightarrow \mu + \sigma\left(\frac{\phi(c^*) - \phi(d^*)}{\phi(d^*) - \phi(c^*)}\right)$$

-Application in Reliability, Queuing. etc. (of Truncated distribution)

25/11/12

Cumulative Dist^n function (c.d.f) :-

$$F_x(x) = P(X \leq x)$$

For discrete Random variable

$$F_X(x) = \sum_{j=0}^{x} P(x=j) \qquad P(X=x) = F_x(x) - F_x(x-1)$$

For continuous ~~Dist~~ R.V

$$F_x(x) = \int_{-\infty}^{x} f(x)\,dx$$

$$f(x) = \frac{d}{dx} F_x(x).$$

Property of F :-

1.) F is non-decreasing fu^n.

2) $F(\infty) = 1$.

3.) $F(-\infty) = 0$

4.) F is right continuous.

Proof :-

(1) Say x is r.v with c.d.f $F_x(x)$

Proof if $x \leq y \rightarrow F_x(x) \leq F_x(y)$

pf :- $A_x = [X \leq x]$, $A_y = [X \leq y]$

$A_x \subseteq A_y$ if $A \subseteq B$

$\Rightarrow P(A_x) \leq P(A_y)$ $P(A) \leq P(B)$.

$P(X \le x) \le P(X \le y)$

$\Rightarrow F_x(x) \le F_x(y)$

(2) $F(\infty) = 1$.

$A_x = \{X \le x\}$

$\{A_n\}$ is a monotic non decreasing sequence.

$\lim_{n \to \infty} P(A_n) = P \lim_{n \to \infty} \{A_n\}$

LHS $F(\infty)$ RHS $= P(\Omega) = 1$.

(3) $F(-\infty) = 0$

$B_x = \{X \le -x\}$

$\{B_n\}$ is a monotonic non-increasing seq.

Applying coutinuity theo.

$\lim_{n \to \infty} P(B_n) = P \lim_{n \to \infty} [B_n] \Rightarrow$ LHS $= F(-\infty)$.

RHS $= P(\phi) = 0$.

(4) $F(x+0) = F(x)$

Define $C_n = \{X \le x + \frac{1}{n}\}$

$\{C_n\}$ is monotonic non $\uparrow$ing. seq.

Applying coutinuity theorem.

$\lim_{n \to \infty} P(C_n) = P \lim_{n \to \infty} [C_n]$.

L.H.S $= F(x+0)$, RHS $\Rightarrow P(X \le x) = F(x)$.

Cor. $F(x-0)$

Define $D_n = \left[x \le x - \frac{1}{n}\right]$

$[D_n]$ is monotonic non-decreasing.

Lt $P(D_n) =$ Plt(D_n)

LHS $= F(x-0)$ RHS $= P(X < x)$.

For continuous R.V. $F(x-0) = F(x)$

But for discreate R.V.

$F(x-0) = P(X<x) \neq P(X \le x)$.

Result: ① if X follows Bin(n,p) then $P(X \le K) = \frac{1}{B(n-K, K+1)} \int_0^q z^{n-K+1} (1-z)^K \cdot dz$.

Proof

R.H.S $\Rightarrow \frac{\lceil(n+1)}{\lceil(n-k)\ \lceil(K+1)} \int_0^q z^{n-K+1} (1-z)^K \cdot dz$. (Int. By parts)

$= \frac{n!}{(n-K-1)!\ K!} \left[(1-z)^K \frac{z^{n-K}}{n-K} \Big|_0^q - \int_0^q K(1-z)^{K-1}(-1) \frac{z^{n-K}}{n-k} dz \right]$

$= \frac{n!}{(n-k)!\ K!} p^K q^{n-K} + \frac{n!}{(n-K)!\ (K-1)!} \int_0^q z^{n-K} (1-z)^{K-1} dz$

$\Rightarrow P(X=K) \frac{n!}{(n-k)!(k)!} \int_0^q z^{n-K} (1-z)^{K-1} dz$.

$\Rightarrow P(X=K) \frac{+ n!}{(n-K)!(K-1)!} \left[(1-z)^{K-1} \frac{z^{n-K+1}}{n-K+1} \Big|_0^q - \int_0^q \frac{z^{n-K+1}}{n-k+1} (K-1)(1-z)^{k-2} \times (-1) dz \right.$

$$\Rightarrow P_{(X=k)} + P_{(X=k-1)} + \frac{n!}{(n-k+1)!(k-2)!}\int_0^q z^{n-k+1}(1-z)^{n-2}\,dz.$$

$$= \sum_{0}^{k} P(X=x) = P(x \le k).$$

Result-2

Let $X \sim P(\lambda)$

$$P(X \le k) = \frac{1}{\Gamma(k+1)}\int_{\lambda}^{\infty} e^{-z}\cdot z^{k}\,dz.$$

Proof

RHS $\frac{1}{k!}\left[z^k \frac{e^{-z}}{(-1)}\Big|_{\lambda}^{\infty} - \int_{\lambda}^{\infty} k\cdot z^{k-1}\frac{e^{-z}}{-1}\,dz\right]$

$$= e^{-\lambda}\frac{\lambda^k}{k!} - \frac{1}{(k-1)!}\int_{\lambda}^{\infty} e^{-z}\cdot z^{k-1}dz.$$

$$= P(X=k) + \frac{1}{(k-1)!}\left[z^{k-1}\frac{e^{-z}}{-1}\Big|_{\infty}^{\lambda} - \int_{\infty}^{\lambda}(k-1)\,z^{k-2}\cdot\frac{e^{-z}}{-1}\,dz\right]$$

$$= P_{(X=k)} + P(X=k-1) + \frac{1}{(k-2)!}\int_{\lambda}^{\infty} e^{-z}\cdot z^{k-2}\cdot dz.$$

$$= P(X=k) + P(X=k-1) + \text{------} + P(X=0)$$

$$= P(X \le k)$$

Derived Distribution:-

Situation

Let x be a R.V. with P.d.f $f(x)$., cdf $F(x)$

Let Y be another R.V

$Y = \phi(x) \Rightarrow x = \phi^{-1}(Y)$ (It is one to one fun^)

What is the P.d.f. of Y.

P.d.F of Y is $g(y)$

$$g(y) = f(\phi^{-1}(y)) \, |J| .$$

$$|J| = \left|\frac{dx}{dy}\right|$$

$$g(y) \cdot dy = f(x) \cdot dx$$

$$g(y) = f(x) \cdot \frac{dx}{dy}$$

$$= f(\phi^{-1}(y)) \left|\frac{dx}{dy}\right|$$ (For Prob -ve value will not be there)

Problem:-

(1) Let $x \sim N(\mu, \sigma^2)$, $Y = a + bx$.

Find the distribution of Y.

Sol^:- $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ $\quad -\infty < x < \infty$

$Y = a + bx \Rightarrow x = \frac{Y-a}{b}$.

$|J| = \left|\frac{dx}{dy}\right| = \frac{1}{b}$.

p.d.f of $y = g(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{\frac{Y-a}{b} - \mu}{\sigma}\right)^2} * \frac{1}{|b|}$.

$$= \frac{1}{\sqrt{2\pi}\,|b|\sigma} e^{\frac{-1}{2}\left(\frac{y-a-b\mu}{b\sigma}\right)^2}$$

$$\boxed{Y \sim N(a+b\mu, b^2\sigma^2)}$$

Ques (2) $X \sim U(0,1)$

$$Y = -\frac{1}{\lambda}\log_e(1-x)$$

Find the dist^n of Y.

p.d.f of x.

$$f(x) = 1 \qquad 0 \le x \le 1.$$

$$x = 1 - e^{-\lambda y}$$

$$|J| = \left|\frac{dx}{dy}\right| = \lambda \cdot e^{-\lambda y}.$$

$$g(y) = \text{pdf of } Y = \lambda \cdot e^{-\lambda y}.$$

$$\boxed{Y \sim \exp(\lambda)}$$

---

(3) $Y = X^c$

$Y \sim \exp(1)$

Find the distribution of X.

Sol^n:- p.d.f of $Y = f(Y) = e^{-Y}$

$$Y = X^c \qquad |J| = cX^{c-1}$$

pdf of X:- $g(x) = e^{-x^c} \cdot c \cdot x^{c-1}$.

X is called weibull dist^n.

Find out μ_r' of Weibull Distribution. $Y = X^c$, $X = Y^{1/c}$

μ_r' for X: $E(X^r) = E(Y^{\frac{r}{c}})$

$$= \int_0^\infty e^{-y} \cdot y^{r/c} \cdot dy.$$

$$= \Gamma\left(\frac{r}{c}+1\right) \quad \text{(gamma)}.$$

Prob-4. Y follows $N(\mu, \sigma^2)$

$$Y = \log_e X.$$

Find the distribution of X.

Soln:-

$$|J| = \frac{dy}{dx} = \frac{1}{x}$$

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2}$$

$$-\infty < y < \infty$$

$$g(x) = f(\log_e x) \cdot \frac{1}{x}$$

$$= \frac{1}{x\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{\log_e x - \mu}{\sigma}\right)^2}, \quad 0 < x < \infty.$$

X is said to follow Log Normal Distⁿ.

Find μ_r' of X.

$$\mu_r' = E(X^r)$$

$$= E(e^{rY})$$

$$= e^{r\mu + \frac{r^2\sigma^2}{2}}$$

For $X \sim N(\mu, \sigma^2)$ $= E(e^{tx}) = e^{t\mu + \frac{t^2\sigma^2}{2}}$

Prob ⑤

Let x be a C.R.V. with pdf $f(x)$, cdf $F(x)$.

$Y = F(x)$.

Find the distribution of Y. $\quad x \to Y$.

Sol^n:- $J = \left|\frac{dx}{dy}\right| = \frac{1}{f(x)}$

p.d.f of Y.

$g(y) = f(x)|J|$

$= f(x) \cdot \frac{1}{f(x)}$

$\Rightarrow 1$. $\quad [0 \le Y \le 1]$.

$\boxed{Y \sim U(0,1)}$

Prob ⑥ $X \sim N(0,1)$ (Not one to one Transformation)

$Y = X^2$

Find the dist^n of Y.

Sol^n:- say c.d.f. of Y is $C_r(y)$

$C_r(y) = P(Y \le y)$

$= P(x^2 \le y)$

$= P(-\sqrt{y} \le x \le \sqrt{y})$

$= \Phi(\sqrt{y}) - \Phi(-\sqrt{y})$

pdf of $Y \Rightarrow g(y) = C_r'(y)$.

characteristic fun:

Let x be a R.V & char. fun of X.

$$\phi_x(t) = E(e^{itx}) \qquad i = \sqrt{-1}$$

Cauchy

$f(x) = \frac{1}{\pi} \frac{1}{1+x^2}$

MGF doesn't exist

For discrete R.V.

$$\phi_x(t) = \sum_x e^{itx} P(x=x).$$

For Cont. R.V.

$$\phi_x(t) = \int_{-\infty}^{\infty} e^{itx} f(x) \cdot dx$$

characteristic fun is having its uniqueness prop.

For each dist^n $\exists$ a unique characteristic fun.

$$\boxed{\phi_x(t) = M_x(it)}$$

Probablity generating fun (PGF):-

- exist for D.R.V only (Discrete Random Var)

$$P_x(t) = E(t^x)$$

$$= \sum_x t^x P(x=x).$$

$$\mu[r] = \frac{d^r}{dt^r} P_x(t) \Big|_{t=1}.$$

$$= E(x(x-1)\cdots(x-r+1))$$

29/11/21

Bivariate Probability Distribution:-

Situation:- when Data is collected on more than one variables, it is a case of multivariate Distribution.

when no. of variable - 2.

⇒ It is called Bivariate Distribution.

Bivarivate Joint Prob. Dist^{n}:-

for discrete variable

Bivjoint Prob. mass function $f_{X,Y}(x, y)$

$$f_{X,Y}(x,y) = P(X=x, Y=y)$$

X \ Y	y_1	y_2	y_3
x_1	$f(x_1, y_1)$		
x_2			
x_3			

Properties of $f(x, y)$

1) $0 \le f(x, y) \le 1 \quad \forall x, y.$

2) $\sum_x \sum_y f(x, y) = 1.$

X	P(X=x)

Typical Biv Prob. Distribution:-

X \ Y	y_1	y_2 ---	y_k	
x_1	P_{11}		P_{1K}	$\Rightarrow \sum_{j=1}^{j=k} P_{1j} = P_{10}$
x_2	P_{2L}		P_{2K} .	$\Rightarrow = P_{20}$
⋮				
x_K .	P_{K1}		P_{KK} .	$\Rightarrow = P_{K0}$
	$\sum_{i=1}^{K} P_{i1}$		$\sum_{i=1}^{K} P_{i2}$	

$P_{ij} = P(X=x_i, Y=y_j) = P(x=x_i, y=y_j)$

Marginal Distribution of X.

X	$P(x=x)$
x_1	$\sum_{j=1}^{K} P_{1j}$
x_2	$\sum P_{2j}$
⋮	
x_K	

Marginal dist^n of X

$$g(x) = P(X=x) = \sum_{y} f(x,y)$$

Marg. Dist^n of Y.

$$h(y) = P(Y=y) = \sum_{x} f(x,y)$$

conditional Dist^n:-

Conditional dist^n of X given that Y=y

$$\boxed{f_1(x/y) = \frac{f(x,y)}{h(y)}}$$

Cond. Dist^n of Y given X=x

$$\boxed{f_2(y/x) = \frac{f(x,y)}{g(x)}}$$

Definition:- X & Y are said to be independent if conditional dist $f_1(x/y) = g(x)$ $\Rightarrow f(x,y) = g(x) \times h(y)$

$f_2(y/x) = h(y)$

X\Y	y_1	y_2	y_3	
x_1	P_{11}	P_{12}	P_{13}	$P_{1\cdot}$
x_2	P_{21}	P_{22}	P_{23}	$P_{2\cdot}$
x_3	P_{31}	P_{32}	P_{33}	$P_{3\cdot}$
	$P_{\cdot 1}$	$P_{\cdot 2}$	$P_{\cdot 3}$	

Bivariate Moment:-

r,s th order Raw Moment.

$$\mu'_{rs} = E(X^r Y^s)$$

$$= \sum_{x,y} \sum x^r y^s p(X=x, Y=y)$$

$$= \sum_{x} \sum_{y} x^r y^s f(x,y)$$

$\mu'_{10} = E(XY^0) = E(X)$ = Marginal expectation of X.

$= E(X) = \sum_x \sum_y x y^0 f(x,y)$

$= \sum_x x \sum_y f(x,y)$

$= \sum_x x g(x) = E(X)$

$\mu'_{11} = E(XY)$

$= \sum_x \sum_y xy f(x,y)$

$\mu'_{01} = E(X^0 Y) = \sum_y y h(y) = E(Y)$

= Marg exp. of Y.

Bivariate Central Moment:-

(r,s)th order central moment

$$\mu_{r,s} = E\left((X-\mu_x)^r (Y-\mu_y)^s\right)$$

$$= \sum_x \sum_y (x-\mu_x)^r (y-\mu_y)^s f(x,y)$$

↓ Joint Prob.

$$\mu'_{1,0} = E(X) = \mu_x$$
$$\mu'_{0,1} = E(Y) = \mu_y$$

$$\mu_{2,0} = E\left((x-\mu_x)^2\right)$$

$$= \sum_x \sum_y (x-\mu_x)^2 f(x,y)$$

$$= \sum_x (x-\mu_x)^2 g(x) = \sigma_x^2 = var(X)$$

$$\mu_{02} = \sigma_y^2 = var(Y)$$

$$\mu_{11} = \sum_x \sum_y (x-\mu_x)(y-\mu_y) f(x,y) = E\left((X-\mu_x)(Y-\mu_y)\right)$$

$$= cov(x,y)$$

$$\text{Corelation co-eff} = \rho_{x,y} = \frac{cov(X,Y)}{\sqrt{var(x) \cdot var(Y)}}$$

$$\boxed{\rho_{x,y} = \frac{\mu_{11}}{\sqrt{\mu_{20}\,\mu_{02}}}}$$

Properties of Expectations:-

Let X be a R.V with finite expectation.

1.) if c is a constant & $P(X=c) = 1$

$$\Rightarrow E(X) = c$$

2) if c is a const.

$$E(cX) = cE(X).$$

3) if $Y = a + bX$.

$$\boxed{E(Y) = a + bE(X)}$$

4) Let X, Y be - Biv R.V.

$$E(X+Y) = \sum_x \sum_y (x+y) f(x,y)$$

$$E(X+Y) = \sum_x \sum_y x f(x,y) + \sum_x \sum_y y \cdot f(x,y)$$

$$= \sum_x x \cdot g(x) + \sum_y y \cdot h(y)$$

$$= E(X) + E(Y)$$

5) $E(aX \pm bY) = aE(X) \pm bE(Y)$

6) $P(X \geq Y) = 1$.

$E(X) \geq E(Y)$

$P(X - Y \geq 0) = 1$

Define $Z = X - Y$.

~~$P(Z) \geq 0$~~. $\Rightarrow \boxed{E(X) \geq E(Y)}$

$P(Z \geq 0) = 1$

$E(Z) \geq 0$

$E(X) - E(Y) \geq 0$ $\Rightarrow \boxed{E(X) \geq E(Y)}$

(VII) $|E(X)| \leq E|X|$

$-|X| \leq X \leq |X|$

(VIII) If X & Y are independent

$$E(XY) = E(X) \cdot E(Y)$$

$$E(XY) = \sum\sum xy\, f(x,y)$$

$$= \sum_x \sum_y xy \cdot g(x) \cdot h(y)$$

$$= \sum_x x \cdot g(x) \sum_y y \cdot h(y)$$

$$= E(X) \cdot E(Y)$$

is the converse True? → Converse is Not Necessarily True.

Ex:-

X \ Y	-1	0	1	
-1	0	1/4	0	1/4
0	1/4	0	1/4	1/2
1	0	1/4	0	1/4
	1/4	1/2	1/4	

$E(XY) = 0 = \sum_x \sum_y x \cdot y \cdot f(x,y)$

$E(X) = 0 = (-1)(\frac{1}{4}) + 0(\frac{1}{2}) + 1(\frac{1}{4}) = 0$

$E(Y) = (-1)(\frac{1}{4}) + 0(\frac{1}{2}) + 1(\frac{1}{4}) = 0$

$E(XY) = E(X) \cdot E(Y)$

But they are not independent

$[f(x,y) \neq g(x) \cdot h(y)]$

Prop. of variance:-

Let X be a R.V., c is a constant

1.) $var(cX) = c^2 var(X)$

Let X, Y be two jointly distributed r.v. then.

(ii) $var(X \pm Y) = \sum\sum (x+y) - E(X+Y))^2 f(x,y)$

$= \sum\sum (x+y - E(X) - E(Y))^2 f(x,y)$

$= E(X - E(X))^2 + E(Y - E(Y))^2 \pm 2E(X - E(X))(Y - E(Y))$

$$\boxed{var(X+Y) = \sigma_x^2 + \sigma_y^2 \pm 2\, cov(X,Y)}$$

(iii) $var(aX \pm b(Y)) = a^2 var(X) + b^2 var(Y) \pm 2ab\, cov(X,Y)$

(IV) $$\boxed{var(X+a) = var(X)}$$

Problem:- ① 2 cards are drawn at random without replacement from a deck containing 3 cards numbered 1, 2 & 3.

Sol Let us define rv (X, Y)

X = # on the first card

Y = # 2nd card.

(i) Find out Joint distribution of X & Y.

(ii) Find out $E(X)$, $var(Y)$

(iii) " conditional expectation $E(X/Y=?)$

X \ Y	1	2	3	
1	0	1/6	1/6	1/3
2	1/6	0	1/6	1/3
3	1/6	1/6	0	1/3
	1/3	1/3	1/3	

X	P(X=x)
1	1/3
2	1/3
3	1/3

Y	P(Y=y)
1	1/3
2	1/3
3	1/3

Cond^n dist^n of $(X/Y=2)$

X	$P(X=x/Y=2)$		
1	$P(X=1/Y=2)$	$= \frac{P(X=1, Y=2)}{P(Y=2)}$	1/2
2	$P(X=2/Y=2)$	=	0
3	$P(X=3/Y=3)$	=	1/2

~~$E(X/Y=2)$~~

$E(X/Y=2)$

$Var(X/Y=2)$

Find out ρ_{XY}

$$\rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} = \frac{-1/3}{\sigma_x \cdot \sigma_y} = -ve$$

$Cov(X,Y) = E(XY) - E(X).E(Y) \Rightarrow \frac{11}{3} - 4 \Rightarrow \left(\frac{-1}{3}\right)$

$= E((X-\mu_x)(Y-\mu_y))$

$E(XY) \Rightarrow \frac{1}{3} + \frac{1}{2} + \frac{1}{3} + 1 + \frac{1}{2} + 1 + 0$

$3 + \frac{2}{3} = 3.67 \Rightarrow \left(\frac{11}{3}\right)$

$E(X) = \frac{1}{3} + \frac{2}{3} + 1 = 2$

$E(Y) = 2.$

Prob (2) 2 identical dice are thrown & the number on there upper most face is noted
Let X = Score on the first dice
Y = Larger score of the two
Find out the joint distribution of X & Y.
Find conditional mean of X., condition var. of Y, corelation co-eff blw X & Y.

Property of Co-relation Co-eff (ρ_{XY})

1.) $-1 \leq \rho \leq 1$
2.) The Absolute value of ρ is independent of scale change & origin change.

Proof :- ρ_{XY} → corel co-eff of X & Y.

$u = a + bx$, $v = c + dY$.

ρ_{uv}, ρ_{XY}.

$$\rho_{XY} = \frac{\text{cov}(X,Y)}{\sqrt{\text{var}(X).\text{var}(Y)}} = \frac{E(XY) - E(X).E(Y)}{\sqrt{\text{var}(X).\text{var}(Y)}}$$

$$\rho_{uv} = \frac{\text{cov}(u,v)}{\sqrt{\text{var}(u).\text{var}(v)}}$$

$\text{var}(u) = b^2 \sigma_x^2$

$\text{var}(v) = b^2 \sigma_y^2$

$\text{cov}(u,v) = E[(u - \mu_u)(v - \mu_v)]$

$\mu_u = a + b\mu_x$

$\mu_v = c + d\mu_y$

$= E[(a + bx - a - b\mu_x)(c + dy - c - d\mu_y)]$

$$= E(b(X-\mu_x)\cdot d(Y-\mu_y))$$

$$= bd\ E[(X-\mu_x)(Y-\mu_y)]$$

$$\rho_{uv} = \frac{bd\ E((X-\mu_x)(Y-\mu_y))}{|b|\sigma_x\,|d|\sigma_y}$$

$$= \frac{bd}{|b||d|}\,\frac{cov(X,Y)}{\sigma_x\cdot\sigma_y} = \frac{b\cdot d}{|b||d|}\,\rho_{X,Y}$$

if b & d are of same sign

then $\boxed{\rho_{uv} = \rho_{X,Y}}$

if b & d are of opp. sign.

then $\boxed{\rho_{uv} = -\rho_{xy}}$

(1) $-1 \le \rho \le 1$

$\rho^2 \le 1$

Proof:- Schwartz's inequality

Let X & Y ~~are~~ be two R.V.

$$\boxed{E^2(XY) \le E(X^2)\,E(Y^2)}$$

Take $h(t) = E(tx-Y)^2 \ge 0$ (as exp of square term)

Find $h'(t) = 0$

$$h(t) = t^2E(X^2) - 2tE(XY) + E(Y^2)$$

$$h'(t) = 2tE(X^2) - 2E(XY) = 0$$

$$\boxed{t_0 = \frac{E(XY)}{E(X^2)}}$$

$$h''(t) = 2E(X^2) \ge 0$$

at $t = t_0$ minimum will attain

$h(t_0) \geq 0$

$$h(t_0) = \frac{E^2(XY)}{E(X^2)} - \frac{2E^2(XY)}{E(X^2)} + E(Y^2)$$

$$= E(Y^2) - \frac{E^2(XY)}{E(X^2)} \geq 0$$

$$\Rightarrow \boxed{E^2(XY) \leq E(X^2)\,E(Y^2)}$$

Consider $\xi = X - \mu_X$ (sai)

$\eta = Y - \mu_Y$.

Apply Schmidtz ineq. on ξ & η

$$E^2(\xi\eta) \leq E(\xi^2)\,E(\eta^2).$$

$$E^2((X-\mu_X)(Y-\mu_Y)) \leq E(X-\mu_X)^2 (Y-\mu_Y)^2$$

$$(cov(X,Y))^2 \leq \sigma_x^2 \sigma_y^2$$

$$\frac{(cov(X,Y))^2}{\sigma_x^2 \sigma_y^2} \leq 1$$

$$\Rightarrow \boxed{\rho_{XY}^2 \leq 1}$$

Books for Bivariate
Johnson & kotz.
Goon Gupta & Das Gupta.

2/12/21

Bivariate Normal Dist^n:-

$f(x,y) \rightarrow$ Joint Biv Pdf for X & Y.

cum. Biv dist^n fu^n. (cummulative)

$$F(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f(x,y)\,.dx.dy$$

Prop of $f(x,y)$

$$\int_x\int_y f(x,y)\,dx.dy = 1$$

Margiual dist^n $g(x) = \int_{-\infty}^{\infty} f(x,y).dy$

$$h(y) = \int_{-\infty}^{\infty} f(x,y).dx$$

$E(x) = \int x.g(x)$

$E(x)$, $var(x)$
$E(y)$, $var(y)$

~~$E(X/Y) =$~~

conditional Dist^n of X/Y

$$f(x/y) = \frac{f(x,y)}{h(y)}$$

$$E(x/y) = \int_{-\infty}^{\infty} x.f(x/y)\,.dx$$

$$var(x/y) = \int_{-\infty}^{\infty} (x - E(x/y))^2 f(x/y).dx$$

Cond^n dist^n of ~~Y/x~~, Y/x.

$$\boxed{g(Y/x) = \frac{f(x,y)}{g(x)}}$$

(X, Y) is Said to follow Biv Normal with Parameters

$\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho$

$$(X,Y) \sim N_2(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$$

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{\frac{-1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2 + \left(\frac{Y-\mu_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{Y-\mu_2}{\sigma_2}\right)\right]}$$

$$(-\infty < x, y < \infty)$$

$$g(x) = \int_{-\infty}^{\infty} f(x,y)\, dy$$

$$= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{\frac{-1}{2(1-\rho^2)}\left(\left(\frac{x-\mu_1}{\sigma_1}\right)^2 + \left(\frac{Y-\mu_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{Y-\mu_2}{\sigma_2}\right)\right)} dy$$

$$u = \frac{Y-\mu_2}{\sigma_2}$$

$$\frac{du}{dy} = \frac{1}{\sigma_2}$$

$$dy = \sigma_2\, du$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} e^{\frac{-1}{2(1-\rho^2)}\left(\left(\frac{x-\mu_1}{\sigma_1}\right)^2 + u^2 - 2u\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\right)} du$$

$$= \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} e^{\frac{-1}{2(1-\rho^2)}\left[u - \rho\left(\frac{x-\mu_1}{\sigma_1}\right)\right]^2 + \left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \rho^2\left(\frac{x-\mu_1}{\sigma_1}\right)^2} du$$

$$(1-\rho^2)\left(\frac{x-\mu_1}{\sigma}\right)^2$$

$$= \frac{e^{\frac{-1}{2}\left(\frac{x-\mu_1}{\sigma}\right)^2}}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} e^{\frac{-1}{2(1-\rho^2)}\left[u - \rho\left(\frac{x-\mu_1}{\sigma_1}\right)\right]^2} . du$$

Transformation $t = \dfrac{u - \rho\left(\frac{x-\mu_1}{\sigma_1}\right)}{\sqrt{1-\rho^2}} \Rightarrow \dfrac{dt}{du} = \dfrac{1}{\sqrt{1-\rho^2}}$

$$g(x) = \frac{e^{\frac{1}{2}\left(\frac{x-\mu_1}{\sigma}\right)^2}}{2\pi\sigma_1\sqrt{1-\rho^2}} \left[\int_{-\infty}^{\infty} e^{-t^2/2} \sqrt{1-\rho^2}\, dt \right]$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma_1} e^{\frac{-1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \cdot dt$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma_1} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2} \qquad -\infty < x < \infty$$

$$\int_{-\infty}^{\infty}\left[\int_{\gamma}^{\infty} f(x,y)\, dy\right] dx = \int_{-\infty}^{\infty} g(x)\, dx$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma_1} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2} \cdot dx$$

$$= 1.$$

$$X \sim N(\mu_1, \sigma_1^2)$$

$$Y \sim N(\mu_2, \sigma_2^2)$$

ρ = Co-Rel co-eff b/w X & Y.

$$f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2+\left(\frac{y-\mu_2}{\sigma_2}\right)^2-2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right)\right]}$$

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2}$$

$$h(y) = \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{1}{2}\left(\frac{y-\mu_2}{\sigma_2}\right)^2}$$

$$f_1(x/y) = \frac{f(x,y)}{h(y)} = \frac{\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2+\left(\frac{y-\mu_2}{\sigma_2}\right)^2-2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right)\right]}}{\frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{1}{2}\left(\frac{y-\mu_2}{\sigma_2}\right)^2}}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2+\left(\frac{y-\mu_2}{\sigma_2}\right)^2-2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right)\right]+\frac{1}{2}\left(\frac{y-\mu_2}{\sigma_2}\right)^2}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2+\left(\frac{y-\mu_2}{\sigma_2}\right)^2-2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right)-(1-\rho^2)\left(\frac{y-\mu_2}{\sigma_2}\right)^2\right]}$$

$$\frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2+\rho^2\left(\frac{y-\mu_2}{\sigma_2}\right)^2-2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right)\right]}$$

$$\frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left(\left(\frac{x-\mu_1}{\sigma_1}\right)-\rho\left(\frac{y-\mu_2}{\sigma_2}\right)\right)^2}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)\sigma_1^2}\left[(x-\mu_1)-\rho\frac{\sigma_1}{\sigma_2}(y-\mu_2)\right]^2}$$

$$x/y \sim N\left(E(x/y), var(x/y)\right)$$

$$E(x/y) = \mu_1 + \rho\frac{\sigma_1}{\sigma_2}(y-\mu_2)$$

$$var(x/y) = \sigma_1^2(1-\rho^2)$$

ρ = co-rel. co-eff of X & Y.

Biv MGF.

$$M_{X,Y}(t_1, t_2) = E\left(e^{t_1X + t_2Y}\right)$$

$$\mu'_{rs} = \frac{d^r}{dt_1^r} \frac{d^s}{dt_2^s} M_{(X,Y)}(t_1, t_2)\Big|_{t_1=0, t_2=0}$$

$$\mu'_{11} = \frac{d^2}{dt_1\, dt_2} M_{(X,Y)}(t_1, t_2)\Big|_{t_1=0, t_2=0}$$

$$\rho_{xy} = \frac{E(X,Y) - E(X).E(Y)}{\sigma_x \cdot \sigma_y} = \rho.$$

$X \sim N(\mu_1, \sigma_1^2)$
$Y \sim N(\mu_2, \sigma_2^2)$
> indep.

Joint distribution of X & Y = Product of their Marginal distribution

when X, Y are indep $f(x,y) = g(x).h(y)$

$$f(x,y) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2} \cdot \frac{1}{\sqrt{2\pi}\,\sigma_2} e^{-\frac{1}{2}\left(\frac{y-\mu_2}{\sigma_2}\right)^2}$$

But if X & Y are ~~non~~ not indep.

$$f(x,y) = f_1(x/y).h(y)$$
$$= f_2(y/x).g(x)$$

Assumption:-

$X \sim N(\mu_1, \sigma_1^2)$

$Y \sim N(\mu_2, \sigma_2^2)$

$\rho \rightarrow$ co-rel co-eff b/w X & Y.

Find out Joint dist^n of X & Y.

Sol^n:- X/Y is a Linear fu^n of Y.

$$E(X/Y) = \alpha + \beta Y.$$

$$\int_{-\infty}^{\infty} x f_1(x/y) \cdot dx = \alpha + \beta Y.$$

$$\iint x f_1(x/y) h(y) dy \cdot dx = \alpha \int_{-\infty}^{\infty} h(y) \cdot dy + \beta \int_{-\infty}^{\infty} y \cdot h(y) dy.$$

or $\iint x \cdot f(x,y) \cdot dy \cdot dx = \alpha + \beta \mu_2$

or $\boxed{\mu_1 = \alpha + \beta \mu_2}$

$$\int x \cdot g(x) \cdot dx = \alpha + \beta \mu_2.$$

or $\mu_1 = \alpha + \beta \mu_2$

$\mu_1 \mu_2 = \alpha \mu_2 + \beta \mu_2^2$ — (1)

$$\iint x \cdot y f_1(x/y) h(y) \cdot dy \cdot dx = \alpha \int y \cdot h(y) dy + \beta \int y^2 h(y) dy$$

or $\iint x \cdot y f(x,y) \cdot dx \cdot dy = \alpha \mu_2 + \beta E(Y^2)$

or $E(XY) = \alpha \mu_2 + \beta E(Y^2)$ — (2).

(2) − (1)

$E(XY) - \mu_1 \mu_2 = \beta [E(Y^2) - \mu_2^2]$ $\Rightarrow$ $\beta = \dfrac{E(XY) - \mu_1 \mu_2}{\sigma_2^2}$

$$\beta = \frac{\rho \sigma_1 \sigma_2}{\sigma_2^2}$$

$$\boxed{\beta = \rho \frac{\sigma_1}{\sigma_2}}$$

$$\alpha = \mu_1 - \beta \mu_2$$

$$\boxed{\alpha = \mu_1 - \rho \frac{\sigma_1}{\sigma_2} \times \mu_2}$$

$$E(X/Y) = \alpha + \beta Y$$

$$= \mu_1 - \rho \frac{\sigma_1}{\sigma_2} \mu_2 + \rho \frac{\sigma_1}{\sigma_2} Y$$

$$= \mu_1 + \rho \frac{\sigma_1}{\sigma_2} (Y - \mu_2)$$

$$var(X/Y) = E\left[X - E(X/Y)\right]^2$$

$$= E\left[(X - \mu_1) - \rho \frac{\sigma_1}{\sigma_2}(Y - \mu_2)\right]^2$$

$$\Rightarrow \left[E\left[(X-\mu_1)^2 - 2\rho \frac{\sigma_1}{\sigma_2}(X-\mu_1)(Y-\mu_2) + \rho^2 \frac{\sigma_1^2}{\sigma_2^2}(Y-\mu_2)^2\right]\right]$$

$$\left(E(X/Y) \Rightarrow \mu_1 + \rho \frac{\sigma_1}{\sigma_2}(Y - \mu_2)\right) \left(cov(X,Y) = \rho \sigma_1 \sigma_2\right.$$

$$\left. var(X/Y) \Rightarrow \sigma_1^2 - 2\rho \frac{\sigma_1}{\sigma_2} cov(X,Y) + \rho^2 \frac{\sigma_1^2}{\sigma_2^2} \sigma_2^2 \right]$$

$$var(X/Y) \Rightarrow \sigma_1^2 - 2\rho \frac{\sigma_1}{\sigma_2} \rho \sigma_1 \sigma_2 + \rho^2 \sigma_1^2$$

$$= \sigma_1^2 - 2\rho^2 \sigma_1^2 + \rho^2 \sigma_1^2 = \sigma_1^2 - \rho^2 \sigma_1^2$$

$$\boxed{var(X/Y) = \sigma_1^2 (1 - \rho^2)}$$

$$X/Y \sim N\left(E(X/Y) = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}(Y - \mu_2), var(X/Y) = \sigma_1^2(1-\rho^2)\right]$$

$$f_1(X/Y) = \frac{1}{\sqrt{2\pi}\, \sigma_1 \sqrt{1-\rho^2}} e^{-\frac{1}{2}\left(\frac{X - \mu_1 - \rho \frac{\sigma_1}{\sigma_2}(Y-\mu_2)}{\sigma_1 \sqrt{1-\rho^2}}\right)^2}$$

$$h(y) = \frac{1}{\sqrt{2\pi}\,\sigma_2} e^{\frac{-1}{2}\left(\frac{y-\mu_2}{\sigma_2}\right)^2}$$

Joint dist^n of x, y.

$$f(x,y) = f_1(x/y) \cdot h(y).$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{\frac{-1}{2}\left(\frac{x-\mu_1-\rho\frac{\sigma_1}{\sigma_2}(y-\mu_2)}{\sigma_1\sqrt{1-\rho^2}}\right)^2} \cdot \frac{1}{\sqrt{2\pi}\,\sigma_2} e^{\frac{-1}{2}\left(\frac{y-\mu_2}{\sigma_2}\right)^2}$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{\frac{-1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \frac{\rho\sigma_1(y-\mu_2)}{\sigma_2}\,2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \rho^2\left(\frac{y-\mu_2}{\sigma_2}\right)^2 + (1-\rho^2)\left(\frac{y-\mu_2}{\sigma_2}\right)^2\right]}$$

$$\Rightarrow \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2\right]}.$$

6/12/21

Discrete Distribution:-

General Rule (Sum of the two discrete R.V).

Let X is a R.V. with PMF $f_1(x_1)$, ~~$f_2(x_1)$~~

X_2 " " $f_2(x_2)$

if dist^n of X_1 & X_2 $\Rightarrow f(x_1, x_2)$

$$Y = X_1 + X_2$$

what is the dist^n of Y.

$$P(Y=y) = \sum_{x_1+x_2=y} \sum f(x_1, x_2)$$

Problem-1 Let $X_i \sim$ iid $Bin(n_2, P)$

$i = 1, 2$

iid → identically and independently dist^{ed}.

$X_1 \sim Bin(n_1, P)$ > indep.

$X_2 \sim Bin(n_2, P)$

Define $Y = X_1 + X_2$

Find the dist^n of Y.

Sol^n:- $P(Y=y) = \sum_{x_1+x_2=y} \sum f(x_1, x_2)$

$$= \sum_{x_1+x_2=y} \sum \binom{n_1}{x} p^{x_1} q^{n_1-x_1} \binom{n_2}{x} \cdot p^{x_2} q^{n_2-x_2}$$

$$= \sum_{x_1+x_2=y} \sum \binom{n_1}{x_1} \binom{n_2}{x_2} p^{x_1+x_2} q^{n_1+n_2-(x_1+x_2)}$$

$$= p^y q^{n_1+n_2-y} \sum_{x_1+x_2=y} \sum \binom{n_2}{n_1} \binom{n_2}{x_2}$$

$$= \binom{n_1+n_2}{y} p^y q^{n_1+n_2-y}.$$

$$\boxed{Y \sim \text{Bin}(n_1+n_2, p)}$$

Generalizing, then

if $X_i \sim$ iid $\text{Bin}(n_i, p)$

$i = 1, 2, \ldots k$

$Y = \sum_{i=1}^{k} X_i$ then $Y \sim \text{Bin}\left(\sum_{i=1}^{k} n_i, p\right)$

Prob-2 Let $X_i \sim$ iid $\text{Poi}(\lambda_i)$

$i = 1, 2.$

$Y = X_1 + X_2$

Find the dist^n of Y.

Sol^n:- $P(Y=y) = \sum\sum_{x_1+x_2=y} f(x_1, x_2)$

$$= \sum\sum_{x_1+x_2=y} e^{-\lambda_1} \frac{\lambda_1^{x_1}}{(x_1)!} e^{-\lambda_2} \frac{\lambda_2^{x_2}}{(x_2)!}$$

$$= \frac{e^{-(\lambda_1+\lambda_2)}}{y!} \sum\sum_{x_1+x_2=y} \frac{y!}{(x_1)!(x_2)!} \lambda_1^{x_1} \cdot \lambda_2^{x_2}$$

$$= \frac{e^{-(\lambda_1+\lambda_2)} (\lambda_1+\lambda_2)^y}{y!}$$

$$\boxed{Y \sim \text{Poi}(\lambda_1+\lambda_2)}$$

Generalizing this.

if $X_i \sim$ iid $\text{Poi}(\lambda_i)$

$i = 1, 2, \ldots k.$

$Y = \sum_{i=1}^{k} X_i$

$$\boxed{Y \sim \text{Poi}\left(\sum_{i=1}^{k} \lambda_i\right)}$$

Prob 3 Let $X_i \overset{iid}{\sim} Bin(n_i, p)$

$Y = X_1 + X_2$

Find the conditional dist^n of X_1 given $Y = y$

Sol^n: $P(X_1 = x_1 / Y = y)$

$$= \frac{P(X_1 = x_1, X_2 = x_2)}{P(Y = y)}$$

$y = x_1 + x_2$

$$= \frac{\binom{n_1}{x_1} p^{x_1} q^{n_1 - x_1} \binom{n_2}{x_2} \cdot p^{x_2} q^{n_2 - x_2}}{\binom{n_1+n_2}{y} p^y \cdot q^{n_1+n_2-y}}$$

$$= \frac{\binom{n_1}{x_1}\binom{n_2}{x_2}}{\binom{n_1+n_2}{y}}$$ $\Rightarrow$ Hypergeometric distribution.

Problem-4. Let $X_i \sim Poi(\lambda_i)$ (iid)

$i = 1, 2.$

$Y = X_1 + X_2$

find the conditional dist^n of X_1 given $Y = y$.

Sol^n: $P(X_1 = x_1 / Y = y) = \frac{P(X_1 = x_1, X_2 = x_2)}{P(Y = y)} \Big| (x_1 + x_2 = y)$

$$= \frac{\frac{e^{-\lambda_1} \lambda_1^{x_1}}{(x_1)!} \cdot \frac{e^{-\lambda_2} \lambda_2^{x_2}}{(x_2)!}}{\frac{e^{-(\lambda_1+\lambda_2)} (\lambda_1+\lambda_2)^y}{y!}}$$

$$= \frac{y!}{x_1!\, x_2!} \frac{\lambda_1^{x_1}\, \lambda_2^{x_2}}{(\lambda_1+\lambda_2)^y}$$

$$= \frac{y!}{x_1!\,(y-x_1)!} \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^{x_1} \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{y-x_1}$$

$$= \frac{y!}{x_1!\,(y-x_1)!} \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^{x_1} \left(1-\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^{y-x_1}$$

$$\sim \text{Bin}\left(y, \frac{\lambda_1}{\lambda_1+\lambda_2}\right)$$

Bivariate Derived Continuous Distribution:-

Let (X, Y) have joint distribution $f(x, y)$

$$(X, Y) \longrightarrow (U, V)$$

$$u = \phi_1(x, y)$$

$$v = \phi_2(x, y)$$

$$\Rightarrow x = \psi_1(u, v)$$

$$y = \psi_2(u, v)$$

what is the joint distribution of U, V.

General Rule:-

$$g(u, v) = f(\psi_1(u, v), \psi_2(u, v))\, |J|.$$

$$|J| = \left|\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}\right|$$

$X \sim f(x)$ for univariate case.

$$Y = \phi(x)$$

$$X = \phi^{-1}(Y)$$

$$g(y) = f(\phi^{-1}(y))\,|J|.$$

$$|J| = \left|\frac{dx}{dy}\right|$$

Problem-1

Let $X \sim N(0,1)$, $Y \sim N(0,1)$ > indep.

Let $u = \frac{X}{Y}$

Find the distribution of u.

Sol^n: Let us assume $v = y$.

$u = \frac{X}{Y}$ $\quad v = y$.

Step-1 Find out joint dist. of X & Y.

$$f(x,y) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} \quad \text{(as independent)}$$

$$= \frac{1}{2\pi} e^{-(x^2+y^2)/2}$$

Step-2 Find out inverse fu^n.

$y = v$.

$x = uv$.

Step-3. Find out $|J|$.

$$|J| = \left| \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \right| = \begin{vmatrix} v & u \\ 0 & 1 \end{vmatrix} = M$$

(Absolute value)

Step-4 Find Joint dist^n of u & v.

$$g(u,v) = \frac{1}{2\pi} e^{-\frac{1}{2}(v^2+u^2v^2)} |v| \Rightarrow \frac{1}{2\pi} e^{-\frac{v^2(1+u^2)}{2}} \cdot |v|$$

Interested to find out dist^n of u.

$$g_1(u) = \int_{-\infty}^{\infty} g(u,v)\,dv$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{v^2}{2}(1+u^2)} \cdot |v|\,dv \quad \text{(Even fo}^n\text{)}$$

$$= \frac{1}{2\pi} \times 2 \int_{0}^{\infty} v \cdot e^{-\frac{v^2(1+u^2)}{2}} \cdot dv.$$

$$v^2 = t$$

$$2v\,dv = dt$$

$$= \frac{1}{2\pi} \int_0^{\infty} \frac{dt}{2} e^{-\frac{(1+u^2)}{2} \cdot t}$$

$$= \frac{1}{2\pi} \frac{\left[e^{-\frac{(1+u^2)}{2} t} \right]_0^{\infty}}{\frac{(1+u^2)}{2}}$$

$$= \frac{-1}{(1+u^2)\pi} \left[0 - 1 \right] \Rightarrow \frac{1}{\pi(1+u^2)}$$

u is called $\Rightarrow$ Cauchy's Distribution)

Ratio of two Normal distribution.

↓

independent std.

Prob-2

Let $X \sim$ Gamma Distribution $(\alpha = 1, P = u)$

$Y \sim$ Gamma Distribution $(\alpha = 1, P = v)$

Or

We can write as $\left.\begin{array}{l} X \sim \text{Gamma}(u) \\ Y \sim \text{Gamma}(v) \end{array}\right\}$ indep.

Define $U = X + Y$, $V = \dfrac{X}{X+Y}$

Find the distribution of U & V.

Sol^n:-

Step-1 Joint dist^n of X & Y.

$$f(x,y) = \frac{1}{\Gamma u \, \Gamma v} e^{-(x+y)} x^{u-1} y^{v-1}$$

$0 < x, y < \infty$.

$$f_1(x) = \frac{1}{\Gamma u} e^{-x} x^{u-1}$$

$$f_2(y) = \frac{1}{\Gamma(v)} e^{-y} y^{v-1}$$

$0 < x, y < \infty$.

Step-2 Inverse.

$X = V(X+Y)$

$\boxed{X = UV}$ $\quad \boxed{Y = U(1-V)}$

Step-3.

$$|J| = \left| \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \right| = \begin{vmatrix} V & u \\ 1-V & -u \end{vmatrix}$$

$$= -uV - u(1-V)$$

$$= -uV - u + uV$$

$$= |-u| = u.$$

Step-4 Joint dist^n of u, v

$$f(x,y) = \frac{1}{\sqrt{u}\,\sqrt{y}} e^{-u} (uv)^{u-1} (u(1-v))^{v-1} \cdot u.$$

$$f = g_1(u) \cdot g_2(v) = \frac{1}{\sqrt{u}\,\sqrt{v}} e^{-u} u^{u+v-1} v^{u-1} (1-v)^{v-1}$$

$\Rightarrow$ u & v are independent

pdf of u: $g_1(u) = c_1 e^{-u} u^{u+v-1}$ $\quad 0 < u < \infty$.

pdf of v: $g_2(v) = c_2 v^{u-1} (1-v)^{v-1}$ $\quad 0 < v < 1$

$$c_1 = \frac{1}{\int_0^\infty e^{-u} u^{u+v-1} \cdot du}$$

$$c_2 = \frac{1}{\int_0^1 v^{u-1} (1-v)^{v-1} dv}$$

$$c_1 = \frac{1}{\Gamma(u+v)}$$

$$c_2 = \frac{1}{\beta(u,v)}$$

$u \rightarrow$ gamma dist^n = Gamma$(u+v)$ $\quad u = x+y$

$v \rightarrow$ beta dist^n. = Beta(u,v) $\quad v = \frac{x}{x+y}$

Prob-3. $x \sim$ Gamma(u), $y \sim$ Gamma(v) $>$ indep.

$u = x+y \quad v = \frac{x}{y}$

Find the dist^n of u & v.

Sol^n:- Joint dist^n of x & y.

$$f(x,y) = \frac{1}{\Gamma(u)\,\Gamma(v)} e^{-(x+y)} x^{u-1} y^{v-1} \qquad 0 < x, y < \infty.$$

$x = vy \quad u = vy + y$ $\Rightarrow$ $y = \frac{u}{1+v}$ $\quad x = \frac{uv}{1+v}$

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$y = \frac{u}{1+v} \qquad x = \frac{uv}{1+v}$$

$$\frac{\partial x}{\partial v} = \frac{u(1+v) - uv}{(1+v)^2}$$

$$= \frac{u}{(1+v)^2}$$

$$= \begin{vmatrix} \frac{v}{1+v} & \frac{u}{(1+v)^2} \\ \frac{1}{1+v} & \frac{-u}{(1+v)^2} \end{vmatrix}$$

$$= \left| \frac{-uv}{(1+v)^3} - \frac{u}{(1+v)^3} \right| = \left| \frac{-u(v+1)}{(1+v)^2} \right|$$

$$\Rightarrow \frac{u}{(1+v)^2}$$

Joint dist^n of u, v

$$g(u,v) = \frac{1}{\sqrt{u}\sqrt{v}} e^{-u} \left(\frac{uv}{1+v}\right)^{u-1} \left(\frac{u}{1+v}\right)^{v-1} \frac{u}{(1+v)^2}$$

$$= \frac{1}{\Gamma(u)\Gamma(v)} e^{-u} u^{(u+v-1)} \frac{v^{u-1}}{(1+v)^{u+v}}$$

$$= g_1(u) \cdot g_2(v)$$

$$g_1(u) = c_1 e^{-u} u^{u+v-1} \qquad 0 < u < \infty$$

$$g_2(v) = c_2 \frac{v^{u-1}}{(1+v)^{u+v}} \qquad 0 < v < \infty.$$

$u \sim \text{Gamma}(u+v)$

$v \sim \text{Beta}_2(u,v)$

$$c_1 = \frac{1}{\Gamma(u+v)}$$

$$c_2 = \frac{1}{\int_0^\infty \frac{v^{u-1}}{(1+v)^{u+v}} dv} = \frac{1}{B(u,v)}$$

Problem-4 Joint dist^n of x & y.

$$g(x,y) = \frac{e^{-(x+y)} \cdot x^3 y^4}{\Gamma 4 \; \Gamma 5}, \quad x, y > 0.$$

Find the pdf of $u = \frac{x}{x+y}$.

Sol^n:-

Assume $v = x+y$.

$x = uv$ $\quad y = v - uv$

$= v(1-u)$

$$|J| = \begin{vmatrix} v & u \\ -v & (1-u) \end{vmatrix} \Rightarrow v(1-u) + uv$$

$$v$$

$$f(u,v) = \frac{|x| e^{-v} (uv)^3 (v(1-u))^4 \cdot v}{\Gamma 4 \; \Gamma 5}$$

$$= \frac{1}{\Gamma 4 \; \Gamma 5} e^{-v} v^8 (1-u)^4 u^3$$

$$= g_1(u) \cdot g_2(v)$$

$$g_1(u) = c_1 u^3 (1-u)^4 \qquad 0 < u < 1.$$

$$c_1 = \frac{1}{\beta(4,5)}$$

Exercise $f(x,y) = c\,x\,y$

$0 < x, y < 1$

Find $c, g(x), E(X/Y)$

Exercise $(X,Y) \sim f(x,y)$

$f(x,y) = c \cdot e^{-8x^2 - 6xy - 18y^2}$

Find out c & ρ_{xy}

Soln:- $f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\, e^{\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2 + \left(\frac{y-\mu_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right)\right]}$

$\mu_1 = 0 \quad \mu_2 = 0.$

Now equate the co-efficient & find c & ρ_{xy}.

Exercise:-

$(X,Y) \sim N_2(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$

$Z = a.X + b.Y.$

Find the dist^n of Z.

$P(Z > 0)$

Soln:-

$X \sim N(\mu_1, \sigma_1^2)$

$Y \sim N(\mu_2, \sigma_2^2)$

} we have derived,

$cov(X,Y) = \rho\sigma_1\sigma_2$

$Z \sim N\left(E(Z), var(Z)\right)$

$E(Z) = a\mu_1 + b\mu_2$

$var(Z) = a^2\sigma_1^2 + b^2\sigma_2^2 + 2(a,b,\rho)\,\mu$

Properties of MGF:-

(1) Let X be R.V with MGF $M_X(t)$.

$$Y = a + bX$$

what is the MGF of Y.

Sol:

$$M_X(t) = E(e^{tX})$$

$$M_Y(t) = E(e^{tY}) = E(e^{t(a+bX)})$$

$$= e^{at} E(e^{tbX})$$

$$= e^{at} M_X(bt)$$

---

(2) Let $X_i \sim$ iid R.V with MGF $M_{X_i}(t)$

$$i = 1, 2, \ldots, k.$$

$$Y = X_1 + X_2 + \cdots + X_k$$

$$M_Y(t) = ?$$

$$M_Y(t) = E(e^{tY}) = E(e^{t\sum_{i=1}^{k} X_i})$$

$$= \prod_{i=1}^{k} E(e^{tX_i})$$

$$= \prod_{i=1}^{k} M_{X_i}(t).$$

Result $X_i \sim Bin(n_i, P)$

$$Y = \sum_{i=1}^{k} X_i$$

Find MGF of Y & identify distribution of Y.

(by unique MGF Property)

Characteristic fu^n - (Properties)

1.) Let X be a R.V with char fu^n $\phi_X(t)$

$$Y = a + bX$$

$$\phi_Y(t) = E(e^{itY})$$

$$= E(e^{it(a+bX)})$$

$$= e^{ita}\, E(e^{itbX})$$

$$= e^{ita}\, \phi_X(bt)$$

2.) Let $X_i \sim$ iid R.V with char ~~eq~~ fu^n $\phi_X(t)$.

$$Y = \sum_{i=1}^{k} X_i$$

$$\phi_Y(t) = E(e^{itY})$$

$$= E\left(e^{it\sum_{i=1}^{k} X_i}\right)$$

$$= \prod_{i=1}^{k} E(e^{itX_i})$$

$$= \prod_{i=1}^{k} \phi_{X_i}(t)$$

Prop. of PGF :-

$$P_X(t) = E(t^X)$$ defined for D.R.V only.

$$Y = a + bX$$

$$P_Y(t) = E(t^{(a+bX)}) = t^a E(t^{bX})$$

$$= t^a P_{bX}(t)$$

$X_i \sim$ iid R.V.

with PGF $P_{X_i}(t)$.

$$Y = \sum_{1}^{k} X_i$$

$$P_Y(t) = E(t^Y)$$

$$= E\left(t^{\sum_{i=1}^{k} X_i}\right)$$

$$= \prod_{i=1}^{k} P_{X_i}(t)$$

(occur only when they are independent).

$X_i \sim$ iid R.V

with PGF $P_{X_i}(t)$.

$$Y = \sum_{1}^{k} X_i$$

$$P_Y(t) = E(t^Y)$$

$$= E\left(t^{\sum_{i=1}^{k} X_i}\right)$$

$$= \prod_{i=1}^{k} P_{X_i}(t)$$

(occur only when they are independent)

9/12/21 **chi-Square Distribution**

χ^2 distribution

χ^2 dist is defined as sum of square of n independent $N(0,1)$

n ↓ d.o.f (degree of freedom)

$$\chi^2 = \sum_{i=1}^{n} u_i^2 \quad \text{where } u_i \sim \text{iid } N(0,1)$$

~~$\chi^2 =$~~

Pdf of χ^2 dist^n:-

MGF of χ^2 with d.o.f 'n'.

$$M_{\chi^2}(t) = E(e^{t\chi^2})$$

$$= E\left(e^{t\sum_{i}^{n} u_i^2}\right)$$

$$= \prod_{i=1}^{n} E(e^{tu_i^2}) \quad \Rightarrow E(e^{tu_i^2}) = \int_{-\infty}^{\infty} e^{tu_i^2} \frac{1}{\sqrt{2\pi}} e^{-u_i^2/2} \, du_i$$

$$= \frac{1}{\sqrt{2\pi}} \times 2 \int_{0}^{\infty} e^{-u_i^2/2(1-2t)} \, du_i$$

$$z = \frac{u_i^2}{2}(1-2t)$$

$$\frac{dz}{du_i} = u_i(1-2t)$$

$$u_i = \sqrt{\frac{2z}{(1-2t)}}$$

$$du_i = \frac{1}{u_i(1-2t)} \cdot dz$$

$$du_i = \frac{\sqrt{(1-2t)}}{\sqrt{2z}\,(1-2t)} \cdot dz$$

Substituting.

$$= \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-z} \frac{dz}{\sqrt{2z(1-2t)}}$$

$$= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{1-2t}} \left(\int_0^\infty e^{-z} \cdot z^{-1/2} \cdot dz \right) \Rightarrow \sqrt{\frac{1}{2}}$$

$$= \frac{1}{\sqrt{1-2t}} = (1-2t)^{-1/2}$$

$$M_{\chi^2}(t) = \prod_{i=1}^{n} E(e^{tu_i^2})$$

$$= (1-2t)^{-n/2}$$

Result $X \sim \text{Gamma}(\alpha, p)$

$$f(x) = \frac{\alpha^p}{\Gamma(p)} e^{-\alpha x} \cdot x^{p-1} \qquad 0 < x < \infty$$

$$M_X(t) = \frac{\alpha^P}{\Gamma(P)} \int_0^\infty e^{-x(\alpha-t)} x^{P-1} \, dx \qquad z = x(\alpha-t)$$

$$= \frac{\alpha^P}{\Gamma(P)} \int_0^\infty e^{-z} \left(\frac{z}{\alpha-t}\right)^{P-1} \frac{dz}{(\alpha-t)}$$

$$= \left(\frac{\alpha}{\alpha-t}\right)^P = \frac{1}{\left(1-\frac{t}{\alpha}\right)^P}$$

$$= \left(1-\frac{t}{\alpha}\right)^{-P}$$

$\chi^2 \sim \text{Gamma}\left(\alpha = \frac{1}{2}, P = \frac{n}{2}\right)$

By using uniquenen Property of MGF.

p.d.f of χ^2

$$f(\chi^2) = \frac{1}{2^{n/2}\Gamma\left(\frac{n}{2}\right)} \cdot e^{-x^2/2} (x^2)^{n/2-1} \qquad 0 < x^2 < \infty.$$

Define $x = \pm\sqrt{x^2}$

pdf of x.

$$f(x) = \frac{1}{2^{n/2}\Gamma\left(\frac{n}{2}\right)} e^{-x^2/2} (x^2)^{\frac{n}{2}-1} |J| \cdot$$

$\boxed{|J| = 2x}$ → kaai

$$= \frac{1}{2^{\frac{n}{2}-1}\Gamma\left(\frac{n}{2}\right)} e^{-x^2/2} x^{n-1} \qquad [0 < x < \infty)$$

Mean & variance of χ^2 dist^n:

$$M_{\chi^2}(t) = (1-2t)^{-n/2}$$

$$\mu_1' = \frac{d M_{\chi^2}(t)}{dt} = -\frac{n}{2}(1-2t)^{-\frac{n}{2}-1}(-2)$$

$$= n(1-2t)^{-\frac{n}{2}-1}$$

$$\boxed{\mu_1' = n} \qquad (t=0)$$

$$\boxed{E(\chi^2) = \mu_1' = n}$$

$$\mu_2' = \frac{d^2 M_{\chi^2}(t)}{dt^2} = n\left(-\frac{n}{2}-1\right)(1-2t)^{-\frac{n}{2}-2}(-2)$$

$$(\mu_2')_{t=0} = n\left(-\frac{n}{2}-1\right)(-2) = 2n\left[\left(\frac{n}{2}\right)+1\right]$$

$$= n(n+2) = n^2+2n$$

$$\text{var} = \mu_2 = \mu_2' - \mu_1'^2$$

$$= n^2 + 2n - n^2$$

$$= 2n$$

$$\boxed{S.d = \sqrt{2n}}$$

Mode of χ^2 dist^n:

$$f(\chi^2) = \frac{1}{2^{n/2}\,\Gamma(\frac{n}{2})}\, e^{-\chi^2/2}\,(\chi^2)^{\frac{n}{2}-1}$$

$$f'(\chi^2) = \frac{1}{2^{n/2}\,\Gamma(\frac{n}{2})}\left[-\frac{1}{2}e^{-\chi^2/2}(\chi^2)^{\frac{n}{2}-1} + \left(\frac{n}{2}-1\right)(\chi^2)^{\frac{n}{2}-2}\, e^{-\chi^2/2}\right] = 0$$

$$-\frac{1}{2}(\chi^2)^{\frac{n}{2}-1} + \left(\frac{n}{2}-1\right)(\chi^2)^{\frac{n}{2}-2} = 0$$

$$\left(\frac{n}{2}-1\right)(\chi^2)^{\frac{n}{2}-2} = \frac{1}{2}(\chi^2)^{\frac{n}{2}-1}$$

$$\chi^2 = 2\left(\frac{n}{2}-1\right)$$

$$= n-2.$$

Mode = $n-2$

$$\text{Skewness} = \frac{\text{Mean} - \text{Mode}}{\text{S.d.}}$$

$$= \frac{n-n+2}{\sqrt{2n}} = \sqrt{\frac{2}{n}} > 0.$$

(Positively skewed)

Characteristics eqⁿ :-

$$\phi_X(t) = \int_0^\infty e^{itx} \frac{1}{2^{n/2}\,\Gamma(\frac{n}{2})} e^{-x^2/2} \cdot (x^2)^{\frac{n}{2}-1} dx$$

$$= M_X(it)$$

$$= (1-2it)^{-n/2} \quad \text{(on integrating)}$$

$$= (1-2it)^{-n/2}$$

* Additivity Property of χ^2 distⁿ :-

Let $\chi_i^2 \sim$ iid χ^2 distⁿ with d.of n_i

$i = 1, 2 \dots k$.

$$Y = \sum_{i=1}^{k} \chi_i^2$$

$$M_Y(t) = E(e^{tx})$$

$$= \prod M_{\chi_i^2}(t)$$

$$= \prod_{i=1}^{n} (1-2t)^{-n_i/2} = (1-2t)^{-\sum_{i=1}^{k} n_i/2}$$

$$= (1-2t)^{-n/2}$$

$$M_{\chi_i^2}(t) = (1-2t)^{-n/2}$$

**

$Y \sim \chi^2$ distⁿ with dof $\sum_{1}^{k} n_i$

Sum of k independent χ^2 with dof n_i is another χ^2 with dof $\sum_{i=1}^{k} n_i$

Problem:-

① $\chi_1^2 \sim \chi^2$ with dof n_1

$\chi_2^2 \sim$ " " n_2

χ_1^2 & χ_2^2 are indep.

$$u = \frac{\chi_1^2}{\chi_2^2}$$

Find the dist^n of u.

Soln:- $u = \frac{\chi_1^2}{\chi_2^2}$, $v = \chi_2^2$

Joint dist^n of χ_1^2 & χ_2^2.

$$f(\chi_1^2, \chi_2^2) = f(\chi_1^2) \cdot f(\chi_2^2)$$

$$= \frac{1}{2^{n_1/2}\left|\frac{n_1}{2}\right.} e^{-\chi_1^2/2} (\chi_1^2)^{\frac{n_1}{2}-1} \times \frac{1}{2^{n_2/2}\left|\frac{n_2}{2}\right.} e^{-\chi_2^2/2} (\chi_2^2)^{\frac{n_2}{2}-1}$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}}\left|\frac{n_1}{2}\right|\left|\frac{n_2}{2}\right.} e^{-(\chi_1^2/2 + \chi_2^2/2)} (\chi_1^2)^{\frac{n_1}{2}-1} (\chi_2^2)^{\frac{n_2}{2}-1}$$

$\chi_2^2 = v$

$\chi_1^2 = uv$

$$|J| = \begin{vmatrix} v & u \\ 0 & 1 \end{vmatrix} = v$$

Joint dist^n of u & v.

$$g(u,v) = \frac{1}{2^{\left(\frac{n_1+n_2}{2}\right)}\left|\frac{n_1}{2}\right|\left|\frac{n_2}{2}\right.} e^{-\frac{v}{2}(1+u)} (uv)^{\frac{n_1}{2}-1} v^{\frac{n_2}{2}-1} \cdot v$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}}\left|\frac{n_1}{2}\right|\left|\frac{n_2}{2}\right.} e^{-\frac{v}{2}(1+u)} u^{\frac{n_1}{2}-1} v^{\frac{n_1}{2}-1+\frac{n_2}{2}}$$

$\neq g_1(u) \cdot g_2(v)$

$$g_1(u) = \int_0^\infty g(u,v)\, dv$$

$$= \frac{u^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \left|\frac{n_1}{2}\right| \left|\frac{n_2}{2}\right|} \int_0^\infty e^{-\frac{v}{2}(1+u)} \cdot v^{\frac{n_1+n_2}{2}-1} \cdot dv$$

$$z = \frac{v(1+u)}{2}$$

$$\frac{dz}{dv} = \frac{1+u}{2} \qquad dv = \left(\frac{2}{1+u}\right) dz \qquad v = \frac{2z}{(1+u)}$$

$$\frac{u^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \left|\frac{n_1}{2}\right| \left|\frac{n_2}{2}\right|} \int_0^\infty e^{-z} \left[\frac{2z}{1+u}\right]^{\left(\frac{n_1+n_2-2}{2}\right)} \left(\frac{2z}{1+u}\right) dz$$

$$\Rightarrow \frac{u^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \left|\frac{n_1}{2}\right| \left|\frac{n_2}{2}\right|} \int_0^\infty e^{-z} \left(\frac{2z}{1+u}\right)^{\left(\frac{n_1+n_2}{2}\right)} dz$$

$$\frac{u^{n_1/2-1}}{2^{\frac{n_1+n_2}{2}} \left|\frac{n_1}{2}\right| \left|\frac{n_2}{2}\right| (1+u)^{\frac{n_1+n_2}{2}}} \int_0^\infty e^{-z} (z)^{\frac{n_1+n_2}{2}} dz$$

$$\frac{u^{n_1/2-1} \left|\frac{n_1+n_2}{2}\right|}{\left|\frac{n_1}{2}\right| \left|\frac{n_2}{2}\right| (1+u)^{\frac{n_1+n_2}{2}}}$$

$$= \frac{1}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \frac{u^{\frac{n_1}{2}-1}}{(1+u)^{\frac{n_1}{2}+\frac{n_2}{2}}} \qquad (0 < u < \infty)$$

$u \Rightarrow$ Beta dist^n of Second kind.

$\sim \text{Beta}_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$

Problem: (2)

$\chi_i^2 \sim$ iid χ^2 with dof n_i

$n = 1, 2$

$U = \dfrac{\chi_1^2}{\chi_1^2 + \chi_2^2}$ Find dist^n of U.

$V = \chi_1^2 + \chi_2^2$

~~$U = \chi_1^2$~~

$$f(\chi_1^2, \chi_2^2) = \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma\frac{n_1}{2} \Gamma\frac{n_2}{2}} e^{-\frac{1}{2}(\chi_1^2+\chi_2^2)} (\chi_1^2)^{\frac{n_1}{2}-1} (\chi_2^2)^{\frac{n_2}{2}-1}$$

~~$= (\chi_1^2)^{\frac{n_1}{2}-1} \cdot (\chi_2^2)$~~

$\chi_1^2 = UV \quad \chi_2^2 = V - UV = V(1-U)$

$$|J| = \begin{vmatrix} V & U \\ -V & 1-U \end{vmatrix} = V - UV + UV = V.$$

Joint dist^n of U & V.

$$g(U,V) = \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma\frac{n_1}{2} \Gamma\frac{n_2}{2}} e^{-\frac{1}{2}(UV + V(1-U))} (UV)^{\frac{n_1}{2}-1} (V(1-U))^{\frac{n_2}{2}-1} \cdot V$$

$$= c\, e^{-\frac{V}{2}} U^{\frac{n_1}{2}-1} (1-U)^{\frac{n_2}{2}-1} \cdot V^{\frac{n_1+n_2}{2}-1}.$$

Dist^n of U.

$$g_1(U) = c_1 U^{\frac{n_1}{2}-1} (1-U)^{\frac{n_2}{2}-1} \quad [0 < U < 1]$$

$$c_1 = \frac{1}{\int_0^1 U^{\frac{n_1}{2}-1} (1-U)^{\frac{n_2}{2}-1} dU}$$

$$g_1(u) = \frac{1}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} u^{\frac{n_1}{2}-1}(1-u)^{\cdots} \qquad C_1 = \frac{1}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)}$$

$$\boxed{0 < u < 1}$$

$$\left[u \sim \text{Beta}_1\left(\frac{n_1}{2}, \frac{n_2}{2}\right)\right] \Rightarrow \text{Dist}^{n} \text{ of } u.$$

$$X_i \sim iid\ N(\mu, \sigma^2)$$

$$Z = \sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma}\right)^2$$

$$Z \sim \chi^2_n$$

Sampling Distributions :-

$$X \sim N(\mu, \sigma^2)$$

Take a random sample of size 'n'.

S1 : $x_1, x_2 - - - - - x_n$ (sample-1)

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$s^2 = \frac{1}{(n-1)} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

S2 : $x_1, x_2 - - - x_n$ (sample-2) → diff from sample-1

$\bar{x} = \checkmark$

$s^2 = \checkmark$

From	S1 →	$\bar{x}_1$	s_1^2	$\tilde{x}_1$	$\breve{x}_1$
From	S2 →	$\bar{x}_2$	s_2^2	$\tilde{x}_2$	$\breve{x}_2$

(Mean, Var, Median, Mode)

different because Sampling fluctuations

Sampling dist^n of $\bar{x}$ & s^2 :-

$x_1, x_2 \dots x_n$ $\quad x \sim N(\mu, \sigma^2)$

$x_i \sim$ iid $N(\mu, \sigma^2)$

$i = 1, 2 \dots n$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2$$

Joint dist^n of $x_1, x_2 \dots x_n$

$$f(x_1, x_2 \dots x_n) = \frac{1}{(\sqrt{2\pi}\sigma)^n} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$(x_1, x_2 \dots x_n) \to (y_1, y_2 \dots y_n)$

~~$y_1 = l_{11}(x_1 - \mu_1) + l_{22}(x_2 - \mu_2)^2 + \dots$~~

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} l_{11} & l_{12} & \dots & l_{1n} \\ l_{21} & l_{22} & \dots & l_{2n} \\ \dots & \dots & \dots & \dots \\ l_{n1} & l_{n2} & \dots & l_{nn} \end{pmatrix} \begin{pmatrix} x_1 - \mu \\ x_2 - \mu \\ \vdots \\ x_n - \mu \end{pmatrix}$$

$$Y_{n\times 1} = l_{n\times n}(x - \mu)_{n\times 1}$$

Helmort'z Transformation :-

$$L = \begin{pmatrix} l_1' \\ l_2' \\ \vdots \\ l_n' \end{pmatrix}$$ Then $\Rightarrow \boxed{L'L = I}$

L is called Helmort'z Transformation matrix

if $l_i' l_i = 1$

$l_i' l_j = 0 \quad \forall \boxed{i \neq j}$

$$l_1' = \left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}} \cdots \frac{1}{\sqrt{n}}\right)$$

$$l_2' = \frac{1}{\sqrt{2}}(1, -1, 0, \cdots 0)$$

$$l_3' = \frac{1}{\sqrt{3}}(1, 1, -2, 0 \cdots 0)$$

$$\vdots$$

$$l_n' = \frac{1}{\sqrt{n(n-1)}}(1, 1, \cdots -(n+1))$$

$$Y_{n\times 1} = L_{n\times n}\left(\underset{\sim}{x} - \underset{\sim}{\mu}\right)_{n\times 1}$$

↳ example of Helwart's Transformation

$$Y_1 = \frac{1}{\sqrt{n}}(x_1 - \mu) + \frac{1}{\sqrt{n}}(x_2 - \mu) + \cdots \frac{1}{\sqrt{n}}(x_n - \mu)$$

$$Y_2 = \frac{1}{\sqrt{2}}(x_1 - \mu) - \frac{1}{\sqrt{2}}(x_2 - \mu) + 0 \cdots + 0.$$

$$\vdots$$

$$Y_n = \left[\frac{1}{\sqrt{n(n-1)}}(x_1 - \mu) + \frac{1}{\sqrt{n(n-1)}}(x_2 - \mu) + \cdots - \frac{(n-1)}{\sqrt{n(n-1)}}(x_n - \mu)\right]$$

$$\underset{\sim}{Y} = L\left(\underset{\sim}{X} - \underset{\sim}{\mu}\right)$$

$$f(x_1, x_2 \cdots x_n) = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n e^{-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$J = J\left(\frac{x_1, x_2, \ldots x_n}{y_1, y_2, \ldots y_n}\right)$$

$$= \frac{1}{J\left(\frac{y_1, y_2 \ldots y_n}{x_1, x_2, \ldots x_n}\right)}$$

$$J\left(\frac{y_1, y_2 \ldots y_n}{x_1, x_2 \ldots x_n}\right) = \left\| \begin{matrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \vdots & & & \\ \frac{\partial y_n}{\partial x_1} & \cdots & & \frac{\partial y_n}{\partial x_n} \end{matrix} \right\|$$

$$= \|L\|$$

$$J = \frac{1}{\|L\|}$$

$$\boxed{J = +1}$$

$$L'L = I$$

$$|L'L| = 1$$

$$|L|^2 = 1$$

$$|L| = \pm 1$$

$$\underset{\sim}{Y} = L(\underset{\sim}{X} - \underset{\sim}{\mu})$$

$$\sum_{i=1}^{n} Y_i^2 = \underset{\sim}{Y}' \underset{\sim}{Y}$$

$$Y = L(\underset{\sim}{X} - \underset{\sim}{\mu})$$

$$= (\underset{\sim}{X} - \underset{\sim}{\mu})' L' L (\underset{\sim}{X} - \underset{\sim}{\mu})$$

$$= (\underset{\sim}{X} - \underset{\sim}{\mu})' I (\underset{\sim}{X} - \underset{\sim}{\mu})$$

$$= \sum_{i=1}^{n} (X_i - \mu)^2$$

$$g(Y_1, Y_2 \ldots Y_n) = \frac{|x| e^{-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{Y_i}{\sigma}\right)^2}}{(\sqrt{2\pi}\sigma)^n}$$

$Y_i \sim$ iid $N(0, \sigma^2)$ $\left[i = 1, 2 \ldots n\right)$

$$Y_1 = \frac{1}{\sqrt{n}}(x_1 - \mu) + \frac{1}{\sqrt{n}}(x_2 - \mu) + \cdots + \frac{1}{\sqrt{n}}(x_n - \mu)$$

$$= \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (x_i - \mu)$$

$$= \frac{n\bar{x} - n\mu}{\sqrt{n}}$$

$$= \sqrt{n}(\bar{x} - \mu)$$

$$\boxed{\bar{x} = \mu + \frac{Y_1}{\sqrt{n}}}$$

$$Y_1 \sim N(0, \sigma^2)$$

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Distribution of s^2 :-

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$(n-1)s^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$= \sum_{i=1}^{n} (x_i - \mu + \mu - \bar{x})^2$$

$$= \sum_{i=1}^{n} \left[(x_i - \mu_i)^2 - 2(x_i - \mu)(\bar{x} - \mu) + (\bar{x} - \mu)^2 \right]$$

$$= \sum_{i=1}^{n} (x_i - \mu)^2 - n(\bar{x} - \mu)^2$$

$$= \sum_{i=1}^{n} Y_i^2 - Y_1^2 = \sum_{i=2}^{n} Y_i^2$$

$$(n-1)\, s^2 = \sum_{i=2}^{n} Y_i^2$$

$$(n-1)\frac{s^2}{\sigma^2} = \sum_{i=2}^{n} \left(\frac{Y_i}{\sigma}\right)^2 \sim \chi^2_{(n-1)} \quad \text{(divide by } \sigma^2\text{)}$$

$$\frac{Y_i}{\sigma} \sim \text{iid } N(0,1)$$

$$\frac{s^2}{\sigma^2} \sim \frac{1}{n-1}\chi^2_{n-1}$$

$$\left(\frac{s}{\sigma} \sim \sqrt{\frac{\chi^2_{n-1}}{n-1}}\right)$$

$\bar{x}$ is a function of Y_1

s^2 is a function of $Y_1, Y_2 \text{---} Y_n$

Y_i's are independent

So Dist^n of $\bar{x}$ & s^2 are independent.

13/12/21 :-

t-distribution

$$t = \frac{X}{\sqrt{Y^2/n}}$$

where $X \sim N(0,1)$

$Y^2 \sim \chi^2$ with dof 'n' (chi-square)

& X & Y^2 are independent

To find out Pdf of t distribution

$$t = \frac{X}{\sqrt{Y^2/n}} \qquad \begin{matrix} X \sim N(0,1) \\ Y^2 \sim \chi^2_n \end{matrix}$$

Pdf of X

$$f_1(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \qquad -\infty < x < \infty.$$

$$f_2(y^2) = \frac{1}{2^{n/2}\Gamma\left(\frac{n}{2}\right)} e^{-y^2/2} (y^2)^{\frac{n}{2}-1}$$

where $0 < y^2 < \infty$

$$y = +\sqrt{y^2}$$

pdf of Y.

$$f_3(y) = \frac{1}{2^{\frac{n}{2}-1}\Gamma\left(\frac{n}{2}\right)} e^{y^2/2} y^{n-1}$$

$$0 < y < \infty.$$

Since X & Y^2 are indep.

X & Y are also indep.

Joint dist^n of X & Y

$$f(x,y) = \frac{1}{\sqrt{2\pi}\, 2^{n/2-1}\, \Gamma\left(\frac{n}{2}\right)} e^{-\frac{1}{2}(x^2+y^2)} \cdot y^{n-1}$$

$$-\infty < x < \infty$$
$$0 < y < \infty$$

$(X,Y) \to (u,\theta)$

$x = u\cos\theta \quad 0 < u < \infty$ [polar Transformation]

$y = u\sin\theta \quad 0 < \theta < \pi$

$$|J| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -u\sin\theta \\ \sin\theta & u\cos\theta \end{vmatrix}$$

$$\Rightarrow u\cos^2\theta + u\sin^2\theta = u$$

Joint distribution of u,θ

$$g(u,\theta) = \frac{1}{\sqrt{2\pi}\, 2^{\frac{n}{2}-1}\, \Gamma\left(\frac{n}{2}\right)} e^{-u^2/2} (u\sin\theta)^{n-1} \cdot u$$

$$= \frac{1}{\sqrt{2\pi}\, 2^{\frac{n}{2}-1}\, \Gamma\left(\frac{n}{2}\right)} e^{-u^2/2}\, u^n \sin^{n-1}\theta$$

$$= g_1(u) \cdot g_2(\theta)$$

$$g_1(u) = c_1 e^{-u^2/2} u^n \qquad 0 < u < \infty$$

$$g_2(\theta) = c_2 \sin^{n-1}\theta \qquad 0 < \theta < \pi$$

$$t = \frac{u\cos\theta}{u\sin\theta/\sqrt{n}}$$

$t = \sqrt{n}\cot\theta$ (not dependent of u dep. on θ only)

t is fu^n of θ only.

$$c_2 = \frac{1}{\int_0^{\pi} \sin^{n-1}\theta \, d\theta} = \frac{1}{\beta\left(\frac{n}{2}, \frac{1}{2}\right)}$$

$$g_2(\theta) = \frac{1}{\beta\left(\frac{n}{2}, \frac{1}{2}\right)} \sin^{n-1}\theta \qquad 0 < \theta < \pi$$

consider the transformation

$\theta \to t$ $\qquad \theta = \cot^{-1}\frac{t}{\sqrt{n}}$

$t = \sqrt{n}\cot\theta$

How to get the dist^n of t

$$|J| = \left|\frac{d\theta}{dt}\right| = \left(\frac{1}{1+\frac{t^2}{n}}\right)\cdot\frac{1}{\sqrt{n}}$$

$$\cot\theta = \frac{t}{\sqrt{n}}$$

$$\sin^{n-1}\theta = \frac{1}{(\operatorname{cosec}\theta)^{n-1}}$$

$$= \frac{1}{(\operatorname{cosec}^2\theta)^{\frac{n-1}{2}}}$$

$$= \frac{1}{(1+\cot^2\theta)^{\frac{n-1}{2}}} = \frac{1}{\left(1+\frac{t^2}{n}\right)^{\frac{n-1}{2}}}$$

$$\left[|J| = \frac{1}{\sqrt{n}\left(1+\frac{t^2}{n}\right)}\right]$$

Pdf of t

$$f(t) = \frac{1}{B\left(\frac{n}{2}, \frac{1}{2}\right) \times \left(1 + \frac{t^2}{n}\right)^{\frac{n-1}{2}}} \times \frac{1}{\sqrt{n}\left(1 + \frac{t^2}{n}\right)}$$

$$-\infty < t < \infty$$

$$= \frac{1}{\sqrt{n}\, B\left(\frac{n}{2}, \frac{1}{2}\right) \left(1 + \frac{t^2}{n}\right)^{\frac{n-1}{2}}}$$

Prop:

1.) $E(t) = 0$

2.) $\text{var}(t) = \frac{n}{n-2}$

3) t dist^n is symmetric dist^n about '0'.

4.) t dist^n is Leptokurtic dist^n.

F-distribution

$$F = \frac{Y_1^2/n_1}{Y_2^2/n_2}$$

where $Y_1^2 \sim \chi^2$ with d.o.f n_1

$Y_2^2 \sim \chi^2$ with dof n_2

& Y_1^2 & Y_2^2 are indep.

$0 < F < \infty$.

Pdf of F dist^n

Since Y_1^2 & Y_2^2 are indep, Y_1 & Y_2 are also indep.

$Y_1 = \sqrt{Y_1^2}$ $\quad Y_2 = \pm\sqrt{Y_2^2}$

Joint dist^n of Y_1 & Y_2

$$f(Y_1, Y_2) = \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}}} e^{-Y_1^2/2} Y_1^{\frac{n_1}{2}-1} \frac{1}{2^{\frac{n_2}{2}-1} \sqrt{\frac{n_2}{2}}} e^{-Y_2^2/2} \cdot Y_2^{n_2-1}$$

Consider the transformation

$(Y_1, Y_2) \to (u, \theta)$ such that

$Y_1 = u\cos\theta \quad 0 < u < \infty$

$Y_2 = u\sin\theta \quad 0 < \theta < \pi/2$

$$|J| = \left| \begin{vmatrix} \frac{\partial Y_1}{\partial u} & \frac{\partial Y_1}{\partial \theta} \\ \frac{\partial Y_2}{\partial u} & \frac{\partial Y_2}{\partial \theta} \end{vmatrix} \right| = \begin{vmatrix} \cos\theta & -u\sin\theta \\ \sin\theta & u\cos\theta \end{vmatrix}$$

$$\Rightarrow u\cos^2\theta + u\sin^2\theta$$

$$\Rightarrow u.$$

Joint dist^n of u & θ:

$$g(u,\theta) = \frac{e^{-u^2/2}(u\cos\theta)^{n_1-1}(u\sin\theta)^{n_2-1}\cdot u}{2^{\frac{n_1+n_2}{2}-2}\left|\frac{n_1}{2}\right|\left|\frac{n_2}{2}\right|}$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}-2}\left|\frac{n_1}{2}\right|\left|\frac{n_2}{2}\right|} e^{-u^2/2} u^{n_1+n_2-1} \cos^{n_1-1}\theta \sin^{n_2-1}\theta$$

$\Rightarrow g_1(u) \cdot g_2(\theta)$

$\Rightarrow u$ & θ are indep.

$$g_1(u) = c_1 e^{-u^2/2} u^{n_1+n_2-1} \qquad 0 < u < \infty$$

$$g_2(\theta) = c_2 \cos^{n_1-1}\theta \sin^{n_2-1}\theta \qquad 0 < \theta < \pi/2$$

$$F = \frac{Y_1^2/n_1}{Y_2^2/n_2} = \frac{u^2\cos^2\theta/n_1}{u^2\sin^2\theta/n_2}$$

$$= \frac{n_2}{n_1}\cot^2\theta.$$

$$\cot^2\theta = \frac{n_1 F}{n_2}$$

$$\left[\theta = \cot^{-1}\sqrt{\frac{n_1 F}{n_2}}\right]$$

$$C_2 = \frac{1}{\int_0^{\pi/2} \cos^{n_1-1}\theta \sin^{n_2-1}\theta \, d\theta}$$

$$= \frac{2}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \quad \text{(Standard Result)}$$

$$g_2(\theta) = \frac{2}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)(\cos\theta)^{n_1-1}(\sin\theta)^{n_2-1}}$$

Consider the transformation

$\theta \to F$

Such that $F = \frac{n_2}{n_1}\cot^2\theta$

$$\cot^2\theta = \frac{n_1 F}{n_2}$$

$$\theta = \cot^{-1}\sqrt{\frac{n_1 F}{n_2}}$$

$$|J| = \left|\frac{d\theta}{dF}\right|$$

$$= \frac{1}{1+\frac{n_1 F}{n_2}} \sqrt{\frac{n_1}{n_2}} \frac{1}{2\sqrt{F}}$$

$$(\cos\theta)^{n_1-1}(\sin\theta)^{n_2-1} = \left(\frac{\cos\theta}{\sin\theta}\right)^{n_1-1} \times (\sin\theta)^{n_1+n_2-2}$$

$$= (\cot^2\theta)^{\frac{n_1-1}{2}}$$

$$\frac{}{(\operatorname{cosec}^2\theta)^{\frac{n_1+n_2-2}{2}}}$$

$$= \frac{(\cot^2\theta)^{\frac{n_1-1}{2}}}{(1+\cot^2\theta)^{\frac{n_1+n_2}{2}-1}}$$

$$= \frac{\left(\frac{n_1 F}{n_2}\right)^{\frac{n_1-1}{2}}}{\left(1+\frac{n_1 F}{n_2}\right)^{\frac{n_1+n_2}{2}-1}}$$

PdF of F

$$f(F) = \left(\frac{2}{B\left(\frac{n_1}{2},\frac{n_2}{2}\right)} \frac{\left(\frac{n_1 F}{n_2}\right)^{\frac{n_1-1}{2}}}{\left(1+\frac{n_1 F}{n_2}\right)^{\frac{n_1+n_2}{2}-1}}\right) \cdot |J|$$

$$|J| = \frac{\sqrt{\frac{n_1}{n_2}}}{2\sqrt{F}\left(1+\frac{n_1}{n_2}F\right)}$$

$$= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1-1}{2}+\frac{1}{2}} F^{\frac{n_1-1}{2}-\frac{1}{2}}}{B\left(\frac{n_1}{2},\frac{n_2}{2}\right)\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}$$

$$f(F) = \frac{\left(\frac{n_1}{n_2}\right)^{n_1/2} F^{\frac{n_1}{2}-1}}{B\left(\frac{n_1}{2},\frac{n_2}{2}\right)\left(1+\frac{n_1 F}{n_2}\right)^{\frac{n_1+n_2}{2}}}$$

$0 < F < \infty$.

F dist^n is a asymmetric dist^n, it can be +ve or -ve skewed, depending on the magnitude of n_1 & n_2.

$F = \frac{Y_1^2/n_1}{Y_2^2/n_2} = \frac{Y_1^2}{Y_2^2/n_2}$

when $n_1 = 1$

$Y_1^2 \sim \chi_1^2$

$Y_2^2 \sim \chi_{n_2}^2$

$F \sim t^2$ distribution with dof n_2

~~If $n_1, n_2 =$~~

$F_{n_1,n_2} = \frac{Y_1^2/n_1}{Y_2^2/n_2}$

$Y_1^2 \sim \chi_{n_1}^2$

$Y_2^2 \sim \chi_{n_2}^2$

$$\frac{1}{F_{n_1,n_2}} = f_{n_2,n_1}$$

~~Application~~ Use of χ^2, t & F in statistical inference

Test of hypothesis.

Parameter : Mean μ

variance σ^2

$H_0: \mu = \mu_0$

Take a sample of size 'n'

$x_1, x_2 ----- x_n$

$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

$$t_o = \frac{(\bar{x} - \mu_0)}{(S/\sqrt{n})} \Rightarrow$$

$x \sim N(\mu, \sigma^2)$

$\bar{x}$ follows $N(\mu, \frac{\sigma^2}{n})$

$$S^2 = \frac{1}{n-1} \Sigma (x_i - \bar{x})^2$$

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

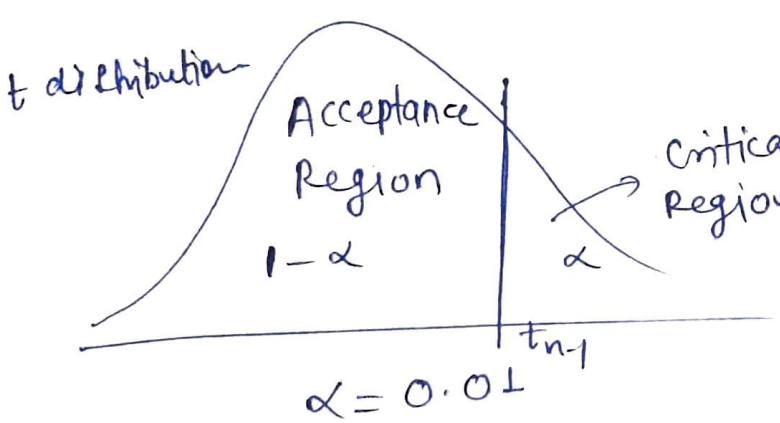
$$\frac{\bar{x} - \mu_0}{S \cdot /\sqrt{n}} = \frac{\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}}{S/\sigma}$$

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$\frac{S^2}{\sigma^2} \sim \frac{\chi^2_{n-1}}{n-1}$$

$$\frac{S}{\sigma} \sim \sqrt{\frac{\chi^2_{n-1}}{n-1}}$$

$$\frac{\bar{x} - \mu_n}{S/\sqrt{n}} \sim \frac{N(0,1)}{\sqrt{\frac{\chi^2_{n-1}}{n-1}}} \sim t_{n-1}$$



$\left(t_o = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}\right)$

↳ if it fall in acceptance Region ✓

if in critical Region ✗

$H_0 : \sigma_1^2 = \sigma_0^2$

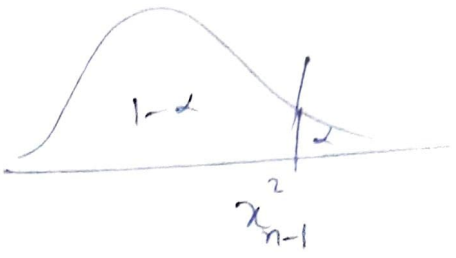
Take a Sample of size n.

Find $S^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$

χ^2 distribution

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$\chi_0^2 = \frac{(n-1)S^2}{\sigma_0^2}$$



$H_0 = \sigma_1^2 = \sigma_2^2 = \sigma^2$

Take Sample of size n_1 from Pop^n 1.

Compute $S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (x_i - \bar{x})^2$

Take a sample of size n_2 from Pop^n 2.

Compute $S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (y_i - \bar{y})^2$

$$F = \frac{S_1^2}{S_2^2}$$

$$\frac{S_1^2}{S_2^2} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1, n_2-1}$$

$$\frac{S_1^2}{\sigma_1^2} \sim \frac{\chi^2_{n_1-1}}{n_1-1} \qquad \frac{S_2^2}{\sigma_2^2} \sim \frac{\chi^2_{n_2-1}}{n_2-1}$$

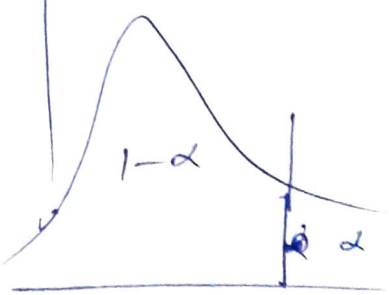
$$\frac{S_1^2}{S_2^2} \sim F_{n_1-1, n_2-1}$$

$x_1, x_2, \ldots, x_n$

$\bar{x} = \frac{1}{n} \sum x_i$

$S^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$

$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$



F_{n_1-1, n_2-1}

$\alpha = 0.01$

$\alpha = 0.05$

16/12/21

Order statistics:-

Let x be a cont. R.V with Pdf $f(x)$, cdf $F(x)$.

Say a sample of size n is taken

Sample of $x_1, x_2 \text{ ---- } x_n$

$X_i \sim$ i.i.d R.V with common Pdf $f(x)$ & cdf $F(x)$.

Consider a situation

After taking the sample only minimum value or max^M value are noted down.

$x_1, x_2 \text{ ------ } x_n$

if it is arranged in order of magnitude

$x_{(1)} \leq x_{(2)} \leq x_{(3)} \leq \text{ ---- } \leq x_{(n)}$

$x_{(1)} = \text{Min}(x_1, \text{ ---- } x_n)$ Range $= R = x_{(n)} - x_{(1)}$

$x_{(n)} = \text{Max}(x_1 \text{ ---- } x_n)$

Due to sampling fluctuations $x_{(1)}, x_{(n)}$ will differ from sample to sample.

$x_1, x_2, x_3 \text{ --- } x_n \rightarrow$ original variable.

$X_{(1)} = \text{Min}(X_1, X_2 \text{ --- } X_n)$

$X_{(n)} = \text{Max}(X_1, X_2 \text{ --- } X_n)$

Order statistics

$X_{(1)}, X_{(2)} \dots X_{(n)}$

Dist^n of $X_{(1)}$ & $X_{(n)}$

$X_i \sim$ iid R.V with common pdf $f(x)$ & cdf $F(x)$

Dist^n of $X_{(n)}$:-

$$\text{cdf of } X_{(n)} = F_{X_{(n)}}(x) = P(X_{(n)} \le x)$$

$$= P(X_{(1)} \le x, X_{(2)} \le x, X_{(3)} \le x \dots X_{(n)} \le x)$$

$$= P(X_1 \le x, X_2 \le x \dots X_n \le x)$$

$$= \prod_{i=1}^{n} P(X_i \le x) \quad \text{(as iid)}$$

$$= \left(F_X(x)\right)^n$$

pdf of $X_{(n)}$

$$f_{X_{(n)}}(x) = n\left(F_X(x)\right)^{n-1} f(x)$$

Dist^n of $X_{(1)}$:-

cdf of $X_{(1)}$ $\quad F_{X_{(1)}}(x) = P(X_{(1)} \leq x)$

$$= 1 - P(X_{(1)} \geq x)$$

$$= 1 - P(X_{(1)} \geq x, X_{(2)} \geq x, \ldots X_n \geq x)$$

$$= 1 - P(X_{(1)} \geq x, X_2 \geq x, \ldots X_n \geq x)$$

$$= 1 - \prod_{i=1}^{n} P(X_i \geq x)$$

$$= 1 - (1 - F(x))^n$$

pdf of $X_{(1)}$

$$f_{X_{(1)}}(x) = n(1 - F(x))^{n-1} f(x)$$

R- chart:-

Dist^n of $R = X_{(n)} - X_{(1)}$

$R = X_{(n)} - X_{(1)}$, $\quad u = X_{(1)}$

Use Bivariate joint dist^n

get dist^n of R.

$E(R) = d_2 \sigma$

Law of Large Number :-

X be a R.V with pdf $f(x)$

$E(x) = \mu$, $Var(X) = \sigma^2$

Sample of size n, $x_1, x_2 \cdots x_n$

$X_i \sim$ iid R.V with common pdf $f(x)$.

$E(X_i) = \mu$, $Var(X_i) = \sigma^2$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i \qquad E(\bar{X}) = \mu.$$

$$Var(\bar{X}) = \frac{\sigma^2}{n}$$

$\bar{X} \to \mu$ as $n \to \infty$.

chebyschev's Lemma :-

Let X be a non neg. RV with finite expectation $E(X) = \mu$, t be a +ve Real no.

$$\boxed{P(X \geq t) \leq \frac{E(X)}{t}}$$

Proof :- Let us assume a R.V Y such that

$$Y = \begin{cases} 0 & \text{if } X < t \\ t & \text{if } X \geq t \end{cases}$$

$$Z = X - Y.$$

$Y \leq X$

$E(Y) \leq E(X)$

$E(Y) = 0 \cdot P(X < t) + t\, P(X \geq t) \leq E(X)$

$$P(X \geq t) \leq \frac{E(X)}{t}$$

chebyshev's inequality:-

Let X be a R.V with mean μ & variance σ^2 & t be a Real no.

$$\boxed{P(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}}$$

Consider a variable

$$Y = (X-\mu)^2$$

Apply cheb. Lemma on Y & t^2

$$P((X-\mu)^2 \geq t^2) \leq \frac{E(X-\mu)^2}{t^2}$$

$$\boxed{P(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}}$$

Implecations:-

cheb inequality provides an upper bound of the prob. that X will deviate from μ by more than t units in terms of σ^2 & t.

Weak Law of Large No. due to kintchine:-

Let $\{X_k\}$ be iid R.V with

$$E(X_k) = \mu, \quad var(X_k) = \sigma^2$$

Let $Z_n = \sum_{j=1}^{n} X_j$

$$E(Z_n) = n\mu$$

$$var(Z_n) = n\sigma^2$$

Jaai $\bar{Z}_n = \dfrac{Z_n - n\mu}{\sigma\sqrt{n}} \to N(0,1)$ as $n \to \infty$

$$X_n = \frac{Z_n}{n}$$

$$E(X_n) = \mu$$

$$var(X_n) = \frac{\sigma^2}{n}$$

$$\frac{X_n - \mu}{\sigma/\sqrt{n}} \to N(0,1) \text{ as } n \to \infty.$$

$$\bar{X}_n = \frac{\bar{\Sigma}_n}{n}$$

$$P[|\bar{X}_n - \mu| \geq t] \to 0 \text{ as } n \to \infty$$

Proof:- $E(\bar{X}_n) = \mu$

$\text{Var}(\bar{X}_n) = \sigma^2/n$

Apply cheb inequality on $\bar{X}_n$

$$P[(\bar{X}_n - \mu) \geq t] \leq \frac{\sigma^2}{nt^2}$$

$$\lim_{n\to\infty} P[(\bar{X}_n - \mu) \geq t] = 0$$

$\bar{X}_n \to \mu$ as $n \to \infty$

 # Bernoulli's theorem:

consider 'n' independent Bernoulli's trials each trial is having two possible outcomes success & failure

success Prob 'p' is Const

& Trials are independent

v = no. of success in n independent trials.

$E(v) = np$

$\text{Var}(v) = npq$

$$P\left[\left|\frac{v}{n} - p\right| \geq t\right] \to 0$$

as $n \to \infty$

Proof: Apply cheb inequality on $\frac{v}{n}$ & t

$$E\left(\frac{v}{n}\right) = p \qquad var\left(\frac{v}{n}\right) = \frac{pq}{n}$$

cheb inequality

$$P\left[\left|\frac{r}{n} - p\right| \geq t\right] \leq \frac{pq}{nt^2}$$

if $n \to \infty$

$$P\left[\left|\frac{r}{n} - p\right| \geq t\right] \to 0.$$

$$\frac{v}{n} \to p \text{ as } n \to \infty$$

First central limit theorem by Lindbergh & Levy

Let $[X_k]$ be a sequence of iid R.V with

mean $E[X_k] = \mu$

$$var[X_k] = \sigma^2$$

(Zaal) $\bar{z}_n = \sum_{j=1}^{n} x_j$

$$E(\bar{z}_n) = \mu$$

$$var(\bar{z}_n) = \sigma^2 n$$

$$Z_n = \frac{\bar{z}_n - n\mu}{\sigma\sqrt{n}} \to N(0,1) \text{ as } n \to \infty.$$

$$\bar{X}_n = \sum_{j=1}^{n} X_j / n \quad \text{(dividing Num}^r\text{, den}^r \text{ by } n\text{)}$$

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \longrightarrow N(0,1) \text{ as } n \to \infty.$$

Proof:- Show that

$$\left(\frac{\sum_{j=1}^{n} x_j - n\mu}{\sigma\sqrt{n}} \right) \longrightarrow N(0,1) \text{ as } n \to \infty$$

$$E(X_j) = \mu$$

$$\text{var}(X_j) = \sigma^2$$

characteristic fun^n of X_j

$$\phi_{X_j}(t) = E(e^{itX_j})$$

$$= \sum_{0}^{\infty} \frac{(it)^r}{r!} \mu_r'$$

$$\phi_{X_j - \mu}(t) = E(e^{it(X_j - \mu)})$$

$$= \sum_{r=0}^{\infty} \frac{(it)^r}{r!} \mu_r \qquad i = \sqrt{-1}$$

$$= 1 + \frac{it}{1!}\mu_1 + \frac{(it)^2}{2!}\mu_2 + \theta\left(\frac{t^3}{3!}\right)$$

θ is containing all higher order terms

$$= 1 + 0 - \frac{t^2\sigma^2}{2} + \theta\left(\frac{t^3}{3!}\right)$$

$$\phi_{\frac{x_j-\mu}{\sigma\sqrt{n}}}(t) = \phi_{x_j-\mu}\left(\frac{t}{\sigma\sqrt{n}}\right)$$

$$= 1 + \frac{t^2}{2}\left(\frac{t}{\sigma\sqrt{n}}\right)^2 + \frac{0}{3!}\left(\frac{t}{\sigma\sqrt{n}}\right)^3$$

$$= 1 - \frac{t^2}{2n} + \frac{0}{3!}\left(\frac{t^3}{\sigma^3 n\sqrt{n}}\right)$$

$$= 1 + \frac{Y}{n}$$

where $Y = -\frac{t^2}{2} + \frac{0}{3!}\left(\frac{t^3}{\sigma^3\sqrt{n}}\right) \to$ as $n \to \infty$, $y \to -\frac{t^2}{2}$

$$\frac{\Sigma x_j - n\mu}{\sigma\sqrt{n}} = \sum_{j=1}^{n} \frac{x_j-\mu}{\sigma\sqrt{n}}$$

$$\phi_{\frac{\Sigma x_j-\mu}{\sigma\sqrt{n}}}(t) = \prod_{j=1}^{n} \phi_{\frac{x_j-\mu}{\sigma\sqrt{n}}}(t)$$

$$= \left(1 + \frac{Y}{n}\right)^n \to e^Y \text{ as } n \to \infty.$$

$$= e^{-t^2/2}$$

as $n \to \infty$

char fun of $\frac{\Sigma(x_j-\mu)}{\sigma\sqrt{n}} = \frac{\Sigma x_j - n\mu}{\sigma\sqrt{n}} \to e^{-t^2/2}$

char fun of $N(0,1)$

Say $X \sim N(0,1)$

$$\phi_X(t) = E(e^{itx})$$

$$= \int_{-\infty}^{\infty} e^{itx} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2 - 2itx)}\, dx = \frac{e^{-t^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-it)^2}\, dx$$

$$= e^{-t^2/2}$$

char. fun of $\frac{\sum X_j - n\mu}{\sigma\sqrt{n}}$ tends to $e^{-t^2/2}$ as $n \to \infty$

$e^{-t^2/2}$ is the char. fun of $N(0,1)$

By the Uniqueness prop of char fun we can say that $\frac{\sum X_j - n\mu}{\sigma\sqrt{n}} \to N(0,1)$ as $n \to \infty$

Cor $\bar{X}_n = \frac{\sum X_j}{n}$

$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \to N(0,1)$ as $n \to \infty$.

(Applicable for all the distribution)

$\frac{\bar{X}_n - E(\bar{X}_n)}{\sigma_{\bar{X}_n}} \to N(0,1)$ as $n \to \infty$

$\mu \pm 3\sigma$ will contain 99.73%

This is the basis of 3σ limit for control chart.

De Moivre's theorem:

Let us consider a seq. of Bernoulli's trials with Succes prob p.

v = no of successes in 'n' independent trials

$E(v) = np$

$Var(v) = npq$

$v \sim Bin(n,p)$

$\frac{v - np}{\sqrt{npq}} \to N(0,1)$ as $n \to \infty$.

Proof:- $v \sim Bin(n,p)$

char fu^n of v

$$\phi_v(t) = E(e^{itx})$$
$$= \sum e^{itx} \binom{n}{x} p^x q^{n-x}$$
$$= (q + pe^{it})^n$$

$X \sim \phi_X(t)$

$Y = a + bX$

$$\phi_Y(t) = E(e^{itY})$$
$$= E(e^{it(a+bX)}) = e^{ita}\phi_X(bt)$$

char fu^n of $\frac{v - np}{\sqrt{npq}}$

$$a = \frac{-np}{\sqrt{npq}} \qquad b = \frac{1}{\sqrt{npq}}$$

$$\phi_{\frac{v-np}{\sqrt{npq}}}(t) = e^{\frac{-itnp}{\sqrt{npq}}} \phi_v\left(\frac{t}{\sqrt{npq}}\right)$$

$$= e^{\frac{-itnp}{\sqrt{npq}}} \left(q + p e^{\frac{it}{\sqrt{npq}}}\right)^n$$

$$= \left(q e^{\frac{-itp}{\sqrt{npq}}} + p e^{\frac{itq}{\sqrt{npq}}}\right)^n$$

Maclaren's expansion:-

For any integer k & real z.

$\exists\ \theta$.

$$e^{iz} = \sum_{j=1}^{k-1} \frac{(iz)^j}{j!} + \frac{\theta z^k}{k!}$$

where θ is a real or complex quantity

$k=3$.

$$e^{\frac{-itb}{\sqrt{npq}}} = 1 + \left(\frac{-itp}{\sqrt{npq}}\right) + \frac{1}{2!}\left(\frac{-itp}{\sqrt{npq}}\right)^2 + \frac{\theta_1}{3!}\left(\frac{-tp}{\sqrt{npq}}\right)^3$$

$$= 1 - \frac{itp}{\sqrt{npq}} - \frac{t^2p^2}{2npq} + \frac{\theta_1}{3!}\left(\frac{-t^3p^3}{(\sqrt{npq})^3}\right)$$

Similarly

$$e^{\frac{itq}{\sqrt{npq}}} = 1 + \frac{itq}{\sqrt{npq}} + \frac{1}{2!}\left(\frac{itq}{\sqrt{npq}}\right)^2 + \frac{\theta_2}{3!}\left(\frac{tq}{\sqrt{npq}}\right)^3$$

$$q \times e^{-itp/\sqrt{npq}} = q - \frac{itpq}{\sqrt{npq}} - \frac{t^2p^2 \cdot q}{2npq} + \frac{\theta_1}{3!}\left(\frac{-t^3p^3}{(\sqrt{npq})^3} \times q\right)$$

$$p \times e^{itq/\sqrt{npq}} = p + \frac{itpq}{\sqrt{npq}} - \frac{t^2q^2 \times p}{2!\,npq} + \frac{\theta_2\, t^3q^3p}{3!\,(\sqrt{npq})^3}$$

$$q e^{\frac{-itq}{\sqrt{npq}}} + p e^{\frac{itq}{\sqrt{npq}}}$$

$$= 1 + 0 - \frac{t^2}{2n}(p+q) + \frac{t^3 p^3}{3!\left(\sqrt{npq}\right)^3}\left(q^2\theta_2 - p^2\theta_1\right)$$

$$= 1 + \frac{Y}{n}$$

$$Y = -\frac{t^2}{2} + o\left(\frac{1}{\sqrt{n}}\right) \to \frac{-t^2}{2} \text{ as } n \to \infty.$$

$$\phi_{\frac{v-np}{\sqrt{npq}}}(t) = \left(q e^{-itp/\sqrt{npq}} + p e^{\frac{itq}{\sqrt{npq}}}\right)^n$$

$$= \left(1 + \frac{Y}{n}\right)^n$$

$$\to e^{Y} \to e^{-t^2/2}$$

as $n \to \infty$.

$e^{-t^2/2}$ is the char fun of $N(0,1)$

By the uniqueness Property of char function we can say that $\frac{v-np}{\sqrt{npq}} \to N(0,1)$ as $n \to \infty$

Cor: $\frac{\frac{v}{n} - p}{\sqrt{\frac{pq}{n}}} \to N(0,1)$ as $n \to \infty$.

we can use 3σ limit to Bin ~~dist~~ variable i.e np chart & p chart.

Definition :-①

A seq. of R.V $\{X_n\}$ is said to be converge in Probability (or weakly) to a R.V X if for $\varepsilon > 0$

$$\lim_{n \to \infty} P[|X_n - X| \geq \varepsilon) = 0$$

$$\Rightarrow X_n \xrightarrow{P} X.$$

Def^n →②

Let $\{X_n\}$ be a seq of rid R.V. whose cdf is F_n, distribution of X is F.

then $\{X_n\}$ is said to converge in distribution if $\lim_{n \to \infty} F_n(x) = F(x)$

→ $\{x_n\}$ is assymptatically distributed on X.

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cheb in-eq

$$P[|X - \mu| \geq t] \leq \frac{\sigma^2}{t^2}$$

1.) ~~tet~~ **Problem**

Let $X \sim$ Poission (m)

Proof that (i) $P\left(X \leq \frac{m}{2}\right) \leq \frac{4}{m}$

(ii) $P(X \geq 2m) \leq \frac{1}{m}$

1) $E(X) = m = \mu$

$Var(X) = m = \sigma^2$

$$\frac{m}{t^2} = \frac{4}{m}$$

$$t^2 = \frac{m^2}{4}$$

$$t = \frac{m}{2}$$

$$P\left(|x-m| \geq \frac{m}{2}\right) \leq \frac{m}{m^2}\left(\frac{4}{m^2}\right)$$

$$P\left(x-m \geq \frac{m}{2}\right) + P\left(x-m \leq -\frac{m}{2}\right) \leq \frac{4}{m}$$

or $P\left(x-m \leq \frac{m}{2}\right) \leq \frac{4}{m}$

or $P\left(x \leq +\frac{m}{2}\right) \leq \frac{4}{m}$

2) $P(x \geq 2m) \leq \frac{1}{m}$

$t = m$

$$P(|x-m| \geq m) \leq \frac{m}{m^2}$$

or $P(x-m \geq m) + P(x-m \leq -m) \leq \frac{1}{m}$

$$P(x-m \geq m) \leq \frac{1}{m}$$

$$P(x \geq 2m) \leq \frac{1}{m}$$